Master Thesis Chapters

A Low-Power, Low-Noise Preamplifier Front-End System for Cochlear Implants

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Chapter 1

Methodology

1.1 Introduction

The main objective of this thesis is the design and optimization of a low-power, low-noise preamplifier suitable for cochlear implant applications. The preamplifier is designed to amplify weak acoustic signals. Specifically, the design targets are low noise, low power dissipation, sufficient gain and bandwidth, low harmonic distortion, high phase margin, low output offset voltage, high common-mode and power-supply rejection, and small silicon area.

The amplifier designed in the present work is based on the conventional folded-cascode CMOS operational amplifier. This topology was selected because of its strong performance in terms of gain, stability, low power consumption, and several other characteristics. Nevertheless, in order to meet the specifications required for its use as a preamplifier in cochlear implants, additional design techniques were applied.

This chapter analyzes in detail the complete design of the Analog Front-End (AFE), which consists of several building blocks. More specifically, the following are examined:

- The preamplifier, which is the main block of the AFE and is intended to amplify the weak input signals received from the microphone.

- The feedback circuit, which ensures system stability and sets the proper gain and operating bandwidth of the amplifier.

- The high-pass filter at the output, which provides a more accurate adjustment of the frequency bandwidth.

- The bias circuit, which biases the amplifier with a stable current and provides the required bias voltages where needed.

- The reference-voltage circuit, which provides a stable reference voltage for biasing the amplifier input.

In this chapter, the different techniques that were applied are examined, and the mathematical calculations required for the circuit design are explained. Particular emphasis is placed on the preamplifier design, since it is the most critical block of the AFE.

1.2 Amplifier Structure

In Figure 1.1, each transistor has the following function:

- Q_1 and Q_2 form the input differential pair.
- Q_3 and Q_4 are cascode common-gate transistors connected to the input common-source transistors.
- Q_5 and Q_6 provide a constant bias current for each branch.
- Q_7 and Q_8, in a composite cascode configuration, operate as source-degeneration resistors.
- Q_9 and Q_{10} operate as the folded-cascode transistors, with the output taken after Q_{10}.
- Q_{11} is used to bias the cascode current mirror without requiring an external bias voltage.
- Q_{12}, Q_{13}, Q_{14}, and Q_{15} form a cascode current mirror that acts as the load and is used to fully exploit the high output resistance provided by the cascode configuration.
- Q_{16}, Q_{17}, Q_{18}, and Q_{19} form two auxiliary amplifiers.
- Q_{20} and Q_{21} provide a constant bias current to the input differential pair.

1.2.1 Design Techniques

The design techniques used are listed below and are discussed in detail in the following sections:

- Input cascode transistors [1].
- Composite transistors operating as source-degeneration resistors [2] [3].
- Active cascode, which includes an auxiliary amplifier [3] [4].
- Flipped current follower, which uses a shunt-shunt feedback technique [5] [3].
- Self-biasing of the cascode current mirror [6] [7] [8].

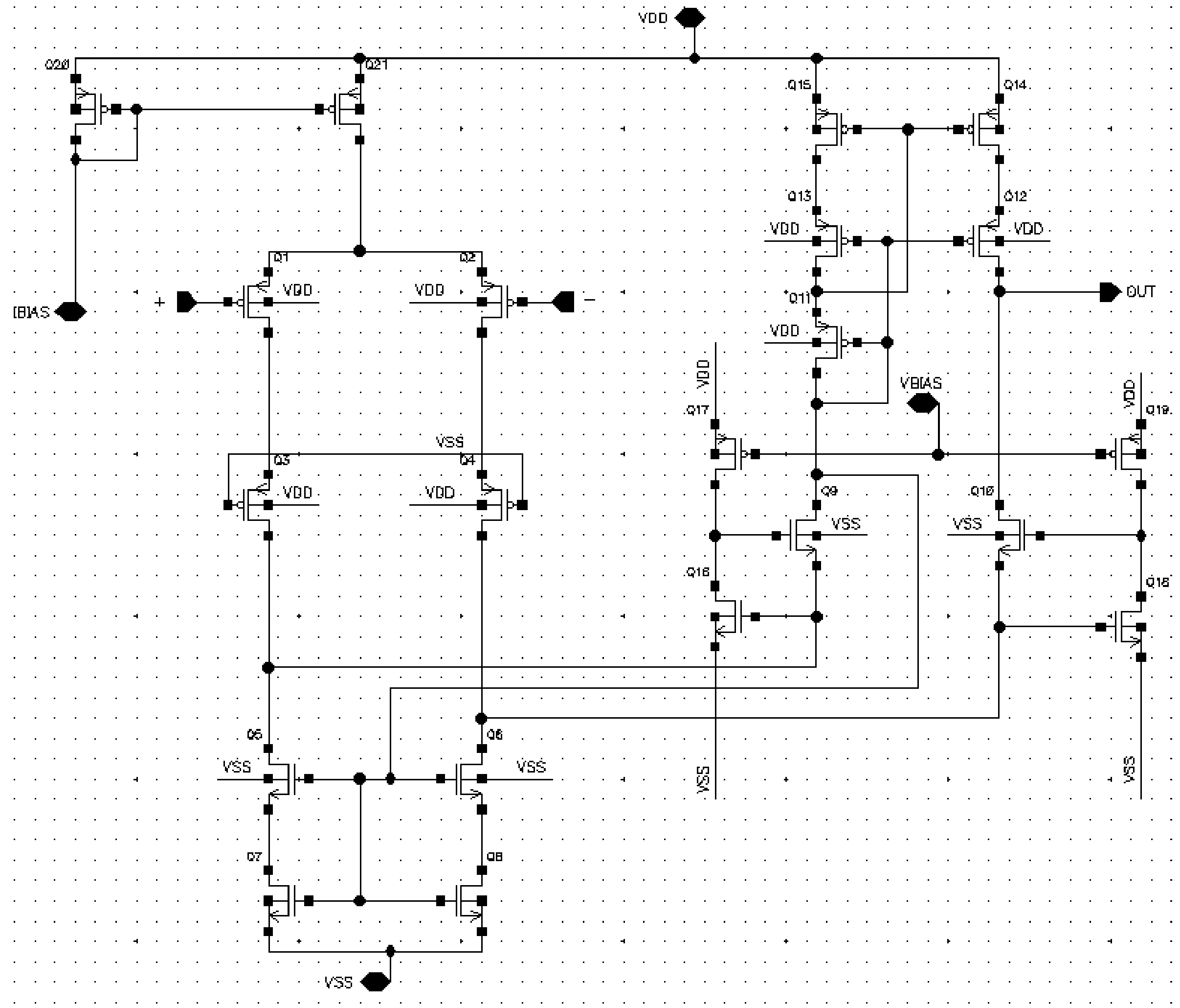


Figure 1.1: Structure of the folded-cascode amplifier of the present work

Conventional Cascode

As shown in Figure 1.2, the Conventional Cascode topology has two stages:

- The input PMOS transistor Q_2 is the amplifying transistor in a common-source configuration, since the AC signal is applied to its gate.
- Q_4 is the cascode transistor connected in a common-gate configuration, since its gate is connected to VSS. It operates as a current buffer, isolating the output of the amplifying transistor.

One of the techniques that helps ensure that the entire differential current generated by the differential input flows through the sources of Q_9 and Q_{10} is the placement of the cascode transistors Q_3 and Q_4 above the input differential-pair transistors. In this way, the resistance seen at the drains of Q_3 and Q_4 is increased, and the differential current is not lost through parasitic paths.

This technique is not required in the conventional folded cascode amplifier, because in that case the current through the output branches is comparable to the input current, which is usually selected to be approximately four times smaller. As a result, the resistance at the sources of the output folded transistors is much smaller than the resistance at the drains of the input transistors, since it is inversely proportional to their transconductance.

In the amplifier of Figure 1.1, the output current is intended to be approximately 25 times smaller than the input-branch current. For this reason, the use of the cascode transistors Q_3 and Q_4 is very important. As will be shown later, other techniques also contribute to achieving this objective, since the cascode transistors alone are not sufficient.

The transconductance of a conventional cascode stage, such as the one in Figure 1.2, is approximately equal to the transconductance of the input transistor. Therefore, $G_m \approx g_{m2}$. The output resistance of Q_2 and Q_4 is approximately:

$$R_{out} = g_{m4} r_{04} r_{o2} \tag{1.1}$$

Considering that Q_1, Q_2, Q_3, and Q_4 operate in the subthreshold region, with Q_1 and Q_2 closer to that region than Q_3 and Q_4, the output resistance can be approximated as follows [9]:

$$R_{out} = \frac{I_{Q4}}{nV_\tau} \frac{1}{\sigma_{Q4} g_{m4}} \frac{1}{\sigma_{Q2} g_{m2}} => \tag{1.2}$$

$$R_{out} = \frac{nV_\tau}{\sigma_{Q2}\sigma_{Q4} I_{SUB}}, \quad g_m = \frac{I_{SUB}}{nV_\tau}, \quad r_0 = \frac{1}{\sigma g_{m2}}, \quad I_{Q2} = I_{Q4} = I_{SUB}, \quad \sigma = \frac{\Delta V_{th}}{\Delta V_D} \tag{1.3}$$

where V_τ is the thermal voltage, approximately 0.026 V, g_m and r_o are the transconductance and output resistance of each transistor, respectively, n is the subthreshold slope factor, and V_{th} is the threshold voltage. The parameter σ is called DIBL (Drain-Induced Barrier Lowering) and describes the reduction of the threshold voltage of a MOSFET when its drain voltage increases. This occurs because a high drain voltage lowers the potential barrier at the source, allowing more current to flow even at low gate voltages. A typical value of σ for a PMOS MOSFET is approximately 0.035.

Composite Transistors and Source Degeneration

Figure 1.3 shows the topology of transistors Q_5 and Q_6, which provide a constant bias current with source-degeneration resistors. In the typical folded-cascode topology shown, these transistors contribute significantly to the noise because of the large currents flowing through them.

By adding source-degeneration resistors, the noise contribution comes mainly from the resistors and can be made much smaller than the noise contribution of MOS transistors operating at the same current level. An additional advantage of the source-degeneration technique is that the resistor noise is mainly thermal, whereas nMOS transistors contribute significantly to flicker noise. Flicker noise appears at low frequencies and is a dominant noise source in amplifiers such as the one designed in the present work.

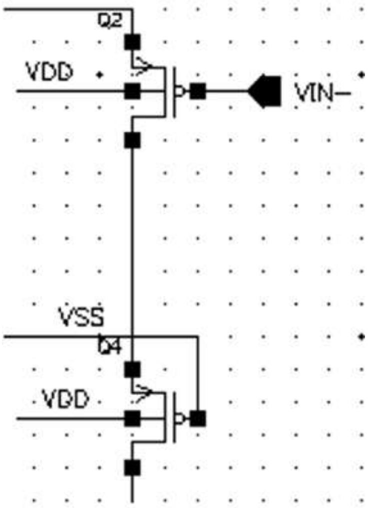


Figure 1.2: Conventional Cascode topology

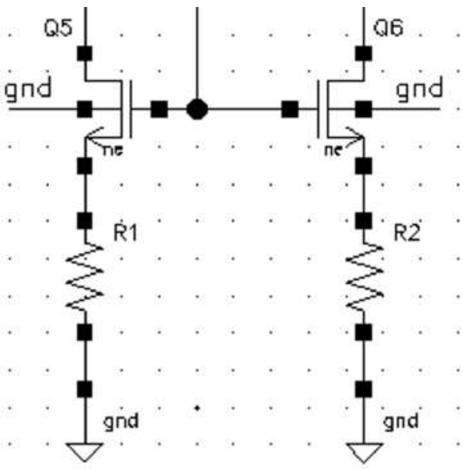


Figure 1.3: Source degeneration using resistors

For source degeneration, the output resistance at the drain is:

$$v_{out} = (i_s + g_m u_s)r_o + R_s i_s => \frac{v_{out}}{i_s} = R_s + r_o + g_m r_o R_s \tag{1.4}$$

For the transconductance, the following expression is obtained [10]:

$$G_m = \frac{g_m r_o}{R_{out}} = \frac{g_m r_o}{R_s + r_o + g_m r_o R_s} \tag{1.5}$$

As shown, this topology provides a considerably high output resistance, which, as will become clear later, is of great importance.

The main disadvantage of the topology in Figure 1.3 is that resistors on the order of hundreds of $k\Omega$ are required, which occupy a large silicon area. A solution to this problem is shown in Figure 1.1, where transistors Q_5, Q_6, Q_7, and Q_8, which form the current source in the lower part of the circuit, implement a composite cascode technique. Transistors Q_5

and Q_6 operate in saturation for improved control of the bias current. Transistors Q_7 and Q_8 operate as signal-dependent source-degeneration resistors and are biased in the linear region. They do not require a large voltage drop across their terminals and therefore do not compromise the circuit voltage headroom. Since the total overdrive voltage of Q_5, Q_7 and Q_6, Q_8 is approximately 260 mV, both transconductance and thermal noise are reduced.

This technique could not be applied directly to the conventional operational amplifier. The reason is that the desired output current is approximately 25 times smaller than the input current, which would require very large widths for the composite transistors [1]. However, the current-source transistors (Q_5, Q_6, Q_7, and Q_8) must have small widths to achieve low noise. This can drive Q_5 and Q_6 into the triode region because of device process variations, resulting in instability. To address this issue, the flipped source follower technique was used and is analyzed in the next section.

Flipped Current Follower

In a common-drain NMOS source follower used as a voltage buffer, the pull-down capability is limited by the bias current source connected to the source of the transistor.

In the circuit of Figure 1.4, the behavior of the source follower is reversed. It provides good pull-down capability, while the pull-up capability depends on the bias current source, which in this case is a cascode current mirror. The pull-down capability of the flipped source follower is due to the very low resistance achieved at the source of transistor Q_9. The reason this technique is called a flipped current follower is that the input signal is a current applied at the source of Q_9. Therefore, because the circuit has current as its input and voltage as its output, the negative feedback is of the shunt-shunt type.

More specifically, the flipped current follower senses the voltage at the drain of Q_9 and applies a feedback current in parallel through transistor Q_5. Transistors Q_9 and Q_5 form a two-pole negative-feedback system. If the loop is opened and a test voltage V_{TEST} is applied to the gate of Q_5, with the output at node B, the open-loop gain is:

$$A_{OL} = \frac{Vout}{V_{TEST}} = -g_{m5} R_{B_{OL}} \tag{1.6}$$

where the open-loop resistance at node B is $R_{B_{OL}} \approx r_{bc} || g_{m9} r_{o9} r_{o5}$, and r_{bc} is the resistance of the cascode current mirror.

The poles of the Q_9–Q_5 system are:

$$\omega_{pB} = \frac{1}{C_B R_{B_{OL}}}, \quad \omega_{pA} = \frac{1}{C_A R_{A_{OL}}} \tag{1.7}$$

where C_B and C_A are the parasitic capacitances at nodes B and A of Q_9 and Q_5, respectively, and the open-loop resistance at the source of Q_9 is $R_{A_{OL}} \approx \frac{(1+(r_{bc}/r_{o9}))}{g_{m9}} || r_{o5}$.

The gain bandwidth is calculated by multiplying the open-loop gain by the dominant pole ω_{pB}, giving $GB = g_{m5}C_B$. According to shunt-shunt feedback theory, the closed-loop resistance at the source of Q_9 is:

$$R_{A_{CL}} = \frac{R_{A_{OL}}}{1 + |A_{OL}|} \approx \frac{\frac{(1+(r_{bc}/r_{o9}))}{g_{m9}}||r_{o5}}{g_{m5}(r_{bc}||g_{m9}r_{o9}r_{o5})} \tag{1.8}$$

In this case, since r_{bc} refers to the output resistance of the cascode current mirror, it can be considered approximately equal to $g_{m9}r_{o9}r_{o5}$, although differences exist because of the additional bias transistor analyzed later. Thus:

$$R_{A_{CL}} \approx \frac{1}{g_{m9}g_{m5}r_{o9}} \tag{1.9}$$

In all cases, the resistance $R_{A_{CL}}$ has a very small value.

For the feedback system to be stable, $\omega_{pB} > 2GB$ must hold. For $r_{bc} \approx g_{m9}r_{o9}r_{o5}$, the following condition is required:

$$\frac{C_B}{C_A} > 2g_{m5}\frac{(r_{o5}/g_{m5}) + r_{o5}^2}{(1/g_{m9}) + 2r_{o5}} \tag{1.10}$$

For the amplifier of Figure 1.1, all parasitic capacitances at the gate, drain, and source of the transistors were taken into account, and the device sizes were selected carefully so that the above condition is satisfied. Therefore, the problem described in the previous section was addressed, since the composite cascode transistors can have small widths without forcing Q_5 and Q_6 out of saturation.

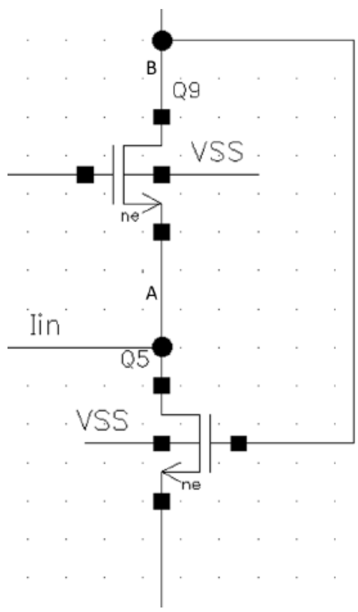


Figure 1.4: Flipped current follower

Active Cascode

The active cascode technique provides additional local gain by using an auxiliary amplifier without relying on multiple cascode transistors. It preserves the positive characteristics of a simple cascode stage while improving gain and output resistance. Therefore, it is particularly popular in modern low-voltage CMOS technologies.

The technique is shown in Figure 1.5. Transistors Q_{16} and Q_{17} form the auxiliary amplifier. Q_{16} is a common-source amplifier, while Q_{17} is the active load. The gate of Q_{17} is biased by VBIAS.

The active cascode technique provides series-series feedback. More specifically, any variation in the current of Q_9 changes the voltage drop across r_{o5}, which in turn directly affects V_{gs9}. The adjusted V_{gs} compensates for the initial current variation, forming a negative-feedback loop. This behavior corresponds to series-series feedback, where the feedback signal is sensed from the output current and mix in series at the input node A. Another local feedback loop is formed by the auxilary amplifier. If the gate voltage of Q_{16} (V_X) increases, the auxiliary-amplifier current increases. The voltage drop across r_{o17} increases, and therefore V_Y decreases. Conversely, if V_X decreases, V_Y increases. Thus, for the auxiliary amplifier, the inverting input can be considered connected to node X, while the non-inverting input is biased by VBIAS.

The additional gain of a conventional cascode topology, as shown in Figure 1.2, comes from the increased output resistance. The active cascode technique shown in Figure 1.5 increases the gain by increasing the effective transconductance of the common-gate transistor. For transistor Q_9:

$$V_{GS9} = V_Y - V_X, \quad V_Y = -AV_X \tag{1.11}$$

where A is the gain of the auxiliary amplifier. Therefore:

$$V_Y - V_X = -(A+1)V_X \tag{1.12}$$

Performing the corresponding calculation for the conventional cascode topology gives $V_Y - V_X = -V_X$. Applying the new V_{GS9} to the small-signal model results in:

$$R_o \approx (A+1)g_{m9}r_{o5}r_{o9}, \quad g_{m9}^{boosted} = g_{m9}(A+1) \tag{1.13}$$

$$G_m = g_{m5}\frac{[(A+1)g_{m9}r_{o9}+1]r_{o5}}{[1+(A+1)g_{m9}r_{o9}]r_{o5}+r_{o9}} \approx g_{m5} \tag{1.14}$$

$$A_{dc} = -G_m R_o \tag{1.15}$$

where g_{m9} is the transconductance without the auxiliary amplifier.

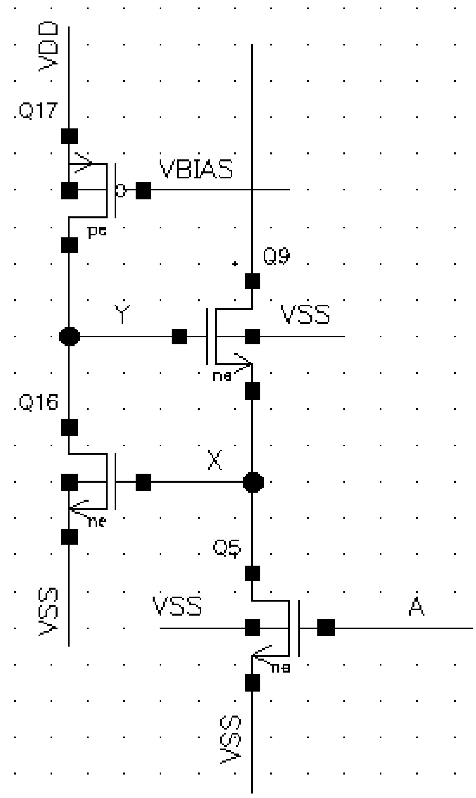


Figure 1.5: Active cascode

Self-Biasing of the Cascode Current Mirror

The conventional Folded Cascode operational amplifier has a limited output-voltage swing. One solution to this problem is the Wide Swing Folded Cascode of Figure ??. However, the drawback of this architecture is that it requires an external bias voltage for biasing transistors Q_5 and Q_6 of the cascode current mirror, increasing the complexity and requirements of the bias circuit.

To address these issues, the circuit of Figure 1.6 was used. It provides improved performance without requiring an external bias voltage. In this design, an additional transistor, Q_{11}, is introduced and operates as a resistor.

All transistors of the cascode current mirror (Q_{12}, Q_{13}, Q_{14}, and Q_{15}), together with the additional bias transistor Q_{11}, operate close to the subthreshold region, which is not desirable due to igher sensitivity to process variations.

Let V_{ON} be the minimum voltage required between the drain and source terminals of a transistor so that it operates in the subthreshold region. Then:

$$|V_{DS,min}| = V_{ON}, \quad I_D = I_0 e^{\frac{|V_{GS}|-|V_{th}|}{nv_T}} \left(1 - e^{-\frac{|V_{DS}|}{v_T}}\right), \quad |V_{GS}| \leq |V_{th}| \tag{1.16}$$

where:

- I_0 is a subthreshold-region coefficient that depends on technology parameters.

- V_{GS} is the gate-source voltage.
- V_{th} is the threshold voltage.
- V_{DS} is the drain-source voltage.
- $V_\tau = \frac{kT}{q}$ is the thermal voltage.
- n is the slope factor in the subthreshold region.

The voltage V_{ON} is approximately equal to $3V\tau$, where $V\tau$ is the thermal voltage. This follows because substituting $|V_{DS}| = 3V\tau$ into the above expression makes the factor $\left(1 - e^{-|V_{DS}|/v_T}\right)$ approximately equal to 1. Thus, I_D depends only on $|V_{GS}|$, resulting in greater stability and lower distortion. If $|V_{DS}|$ is reduced further, the term $e^{-|V_{DS}|/v_T}$ begins to increase and $\left(1 - e^{-|V_{DS}|/v_T}\right) \neq 1$.

Therefore:

$$V_{SD15} = V_{DD} - V_{D15} \geq 3V\tau \Rightarrow V_{D15,max} = V_{DD} - 3V\tau, \tag{1.17}$$

$$V_{D13,max} = V_{DD} - 6V\tau, \tag{1.18}$$

$$V_{D11,max} = V_{DD} - 9V\tau, \tag{1.19}$$

$$V_{D12,max} = V_{DD} - 6V\tau \tag{1.20}$$

Consequently, the amplifier output range before distortion is approximately up to $V_{DD} - 6V\tau$, while the current-mirror input range is up to $V_{DD} - 9V\tau$.

In addition, the acceptable gate-voltage values, considering $|V_{GS}| \leq |V_{th}|$ so that the transistors remain in subthreshold operation, are calculated as follows:

$$V_{G15,max} = V_{D13,max} = V_{DD} - 6V\tau, \tag{1.21}$$

$$V_{G15,min} = V_{D13,min} = V_{DD} - |V_{th}|, \quad \text{since } V_{S15} - V_{G15} \leq |V_{th}|, \tag{1.22}$$

$$V_{G13,max} = V_{D11,max} = V_{DD} - 9V\tau, \tag{1.23}$$

$$V_{G13,min} = V_{D11,min} = V_{DD} - 2|V_{th}|, \quad \text{since } V_{S13} - V_{G13} \leq |V_{th}|. \tag{1.24}$$

The Self-Biased Cascode Current Mirror (SBCCM) shown in Figure 1.6, in addition to providing the bias voltage to the gates of Q_{13} and Q_{14}, also leads to a large output-voltage range before distortion. This reduces the required supply voltage, making it ideal for low-power amplifiers. Furthermore, the SBCCM provides the advantage of high output resistance because of the cascode configuration. Therefore, the output resistance is:

$$r_{out} \approx r_{o14} r_{o12} g_{m12} \tag{1.25}$$

where g_{m12} is the small-signal transconductance of Q_{12} and r_{oi} is the small-signal output resistance of Q_i, $i = 12, 14$.

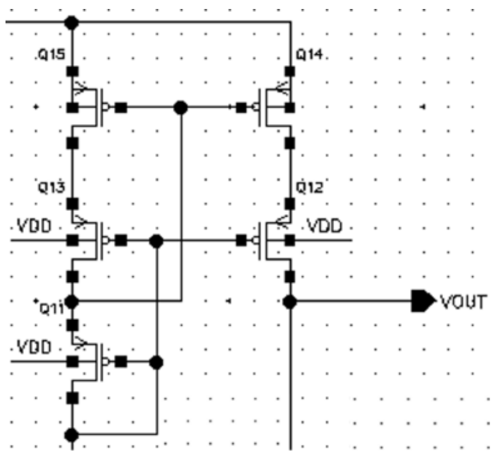


Figure 1.6: Self-biasing of the cascode current mirror

1.3 Feedback System

The feedback system used is shown in Figure 1.7. It is called a Capacitively Coupled Instrumentation Amplifier (CCIA) and is widely used for biomedical preamplifiers. Its transfer function can be approximated as follows [11]:

$$H(s) = \frac{V_{out}}{V_{in+} - V_{in-}} = -\frac{C_1}{C_3}\frac{1}{1 + s\frac{C_1 C_5}{C_3 G_m}}\frac{sR_f C_3}{1 + sR_f C_3} \tag{1.26}$$

where R_f is the pseudo-resistance created by transistors M_1 and M_2 and is equal to the pseudo-resistance of M_3 and M_4. In addition, $C_1 = C_2$ and $C_3 = C_4$. At the output, a high-pass filter is used with transfer function $(sRC_6)/(1 + sRC_6)$. The resistance R is the resistance formed by M_5 and M_6. The overall system has two zeros, $s^2 = 0$, and three poles:

$$s = -\frac{1}{R_f C_3}, \quad s = -\frac{C_3 G_m}{C_1 C_5}, \quad s = -\frac{1}{RC_6}$$

The gain A_d of the amplifier is set by the ratio C_1/C_3 and was selected close to 40dB, a value consistent with the desired amplification before processing in the sound processing unit. This high gain also helps reduce the input-referred noise of the circuit and of the following stages. The bandwidth is given by $BW = \frac{G_m}{2\pi A_d C_5}$. Increasing C_5 reduces the unity-gain frequency and increases the phase margin. Conversely, a small capacitive load increases the amplifier bandwidth but reduces its phase margin. Therefore, the capacitive load C_5 contributes to amplifier stability and hence to frequency compensation.

As shown in Figure 1.7, a high-pass filter is used in the feedback path with cutoff frequency $f = \frac{1}{2\pi R_f C_3}$. This high-pass filter is formed by the pseudo-resistance R_f and the feedback capacitor C_3. The value of R_f is usually selected in the range of $G\Omega$ to $T\Omega$, depending on the application. It also allows the amplifier input to be biased.

The output high-pass filter, consisting of C_6 and R, is used to adjust the bandwidth more accurately; the desired lower cutoff is approximately 20 Hz. In the feedback path, the high-pass filter has a cutoff frequency close to 2 Hz.

The threshold-voltage expression for a MOSFET is:

$$V_{th} = V_{T0} + \gamma \left(\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F} \right) \tag{1.27}$$

where:

- V_{T0} is the threshold voltage for zero source-bulk voltage.
- γ is a body-effect coefficient.
- V_{SB} is the source-bulk voltage.
- ϕ_F is the Fermi potential.

Pseudo-resistors implemented with low-threshold transistors operating in the subthreshold region, as in this design, usually exhibit significant distortion because of the large output swing of the amplifier. As a result, the value of R_f can vary substantially [3]. Therefore, the value of V_{th} should be as large as possible. To achieve this, the bodies of the transistors were connected to the supply for PMOS devices and to ground for NMOS devices. This increased V_{th} by approximately 0.2 V. The drawback is that the threshold voltage now depends on the source voltage of the transistors, making the resistance nonlinear. To address this, one PMOS and one NMOS were connected in series, so that each partially cancels the distortion of the other. Simulations related to this approach are included in Appendix A.

As shown in Figure 1.7, a high-pass filter is used at the output to set the bandwidth. This filter is tuned by two bias voltages, VBIAS1 and VBIAS2, so that the desired pole at 20 Hz is obtained. Since all transistors used as resistors operate very close to cutoff, their resistance depends more on the gate voltage than on their width and length. In addition, a tunable pseudo-resistor exhibits distortion when the voltage across it varies [12]. The reason this output filter was used instead of setting the feedback high-pass filter to the desired frequency is the distortion introduced when the transistors are not diode-connected but are controlled by an external voltage. In the output filter, however, by selecting C_6 so that the required resistance is small, approximately $4G\Omega$, compared with the resistance that would be required in the feedback system, approximately $120G\Omega$, the harmonic distortion decreases as the value of R decreases [13]. Simulations showed that, for the same input-signal values, the harmonic distortion at the output decreased from 8% to less than 1% after introducing the output filter.

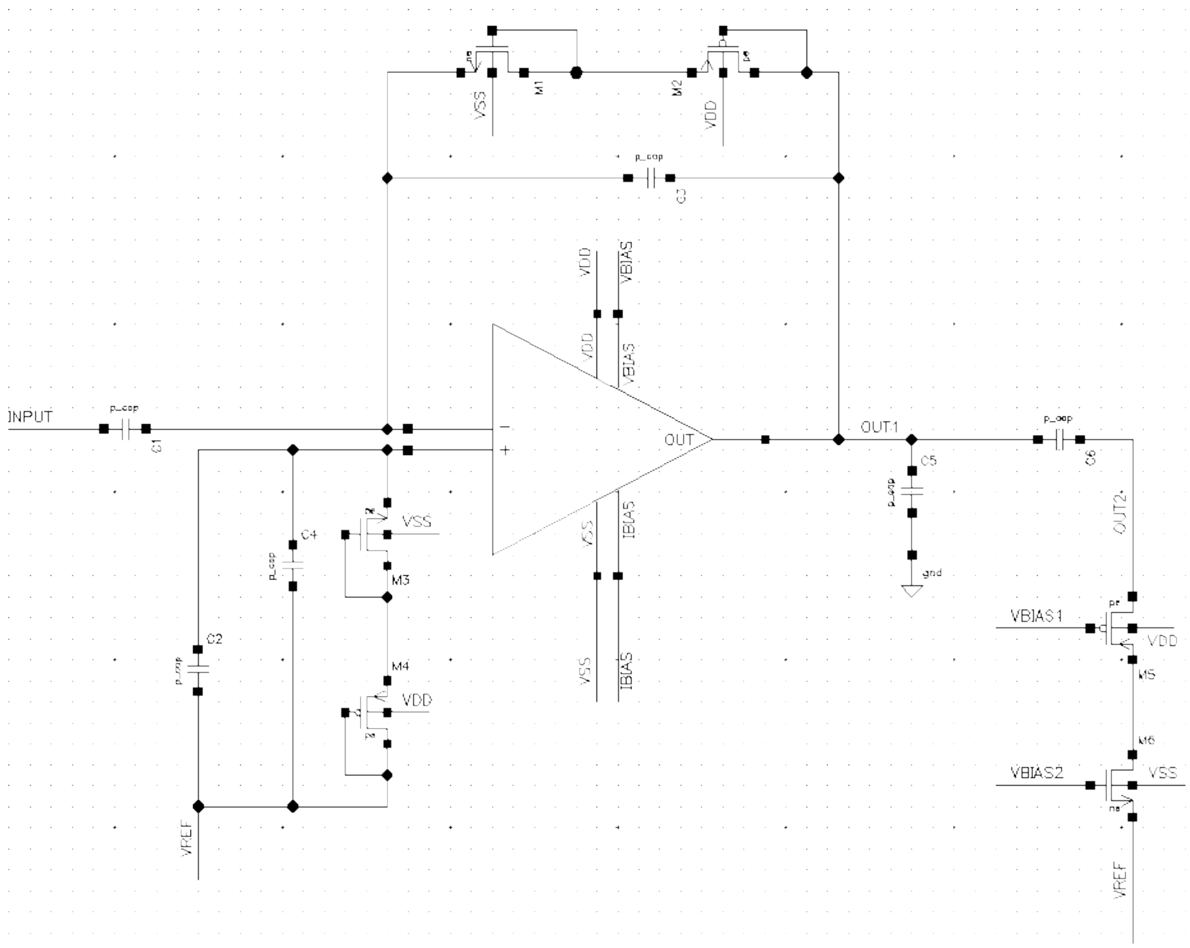


Figure 1.7: Feedback system and high-pass filter at the output

1.4 Theoretical Analysis of Key Specifications

This section presents the theoretical calculations of the specifications. The supply values, transistor dimensions, and operating regions are shown in Tables 1.1 and 1.2. They refer to the amplifier of Figure 1.1 and to the feedback circuit of Figure 1.7.

Table 1.1: Parameters

Parameters	Values
V_{DD}	$1.8V$
C_1, C_2	$7pF$
C_3, C_4	$68fF$
C_5	$650fF$
C_6	$2pF$

1.4.1 Transconductance and Power-Consumption Analysis

The total transconductance of a folded-cascode amplifier should be equal to the transconductance of the input transistors in order to be maximized. Therefore, $G_m \approx g_{m1}$ should

Table 1.2: Amplifier transistor dimensions

Transistors	W/L	Operating region
M_1, M_3	(1/1)	subthreshold
M_2, M_4	(1/0.18)	subthreshold
M_5	(5/0.18)	subthreshold
M_6	(1/1)	subthreshold
Q_1, Q_2	$(22.5/1) \times 8$	subthreshold
Q_3, Q_4	(5/0.3)	subthreshold
Q_5, Q_6	$(3/20) \times 2$	saturation
Q_7, Q_8	(3/20)	linear
Q_9, Q_{10}	(1/0.18)	subthreshold
Q_{11}	(10/0.5)	subthreshold
Q_{12}, Q_{13}	$(2.5/0.18) \times 2$	subthreshold
Q_{14}, Q_{15}	$(5/0.18) \times 2$	subthreshold
Q_{16}, Q_{18}	(4/0.18)	subthreshold
Q_{17}, Q_{19}	(0.5/15)	saturation

hold, as shown in Figure 1.1.

In the present design, the transconductance of the amplifier is given by [1]:

$$G_m = g_{m1} \left(\frac{G_{S9}}{G_{S9} + G_{D5}} \right) \left(\frac{G_{S3} r_{o1}}{1 + G_{S3} r_{o1}} \right) \tag{1.28}$$

where g_{m1} is the transconductance of the input transistor, G_{S9} is the conductance looking into the source of Q_9, G_{D5} is the conductance looking into the drain of Q_5, G_{S3} is the conductance looking into the source of Q_3, and r_{o1} is the output resistance of the input transistor.

Thus, for $G_m \approx g_{m1}$ to hold, the conditions $G_{S9} >> G_{D5}$ and $G_{S3} r_{o1} >> 1$ must be satisfied. To calculate G_{S9}, the closed-loop resistance $R_{S9,CL}$ of the feedback loop must be calculated. According to the closed-loop resistance formula for shunt-shunt feedback [5]:

$$R_{S9,CL} = \frac{R_{S9,OL}}{1 + |A_{OL}|} \tag{1.29}$$

where $|A_{OL}|$ and $R_{S9,OL}$ are the open-loop gain and open-loop resistance, respectively. The resistance $R_{S9,OL}$ is calculated from the parallel combination of the resistance of the composite transistor (R_2), the input cascode transistors (R_1), and the output transistor together with the cascode current mirror (R_3). These quantities were calculated in previous sections:

$$R_1 \approx g_{m3} r_{o3} r_{o1} \tag{1.30}$$

$$R_2 = r_{o7} + r_{o5} + g_{m5} r_{o5} r_{o7} \tag{1.31}$$

$$R_3 \approx \frac{(1 + (r_{bc}/r_{o9}))}{g_{m9}} \approx \frac{(r_{o9} + r_{o11} + (1/g_{m15}))}{g_{m9} r_{o9}} \tag{1.32}$$

Regarding R_3, r_{bc} is the input resistance of the current mirror that uses the bias transistor Q_{11} and can be approximated by $(1/g_{m15}) + r_{o11}$ [8]. Therefore:

$$R_{S9,OL} = R_3||R_2||R_1 = \frac{(r_{o9} + r_{o11} + (1/g_{m15}))}{g_{m9}r_{o9}}||(r_{o7}+r_{o5}+g_{m5}r_{o5}r_{o7})||(g_{m1}r_{o3}r_{o1}) \quad (1.33)$$

The open-loop gain is $g_{m,comp}R_{out}$, where $g_{m,comp}$ and R_{out} are the transconductance of the composite transistor and the total output resistance, respectively. Because of the active cascode technique, R_{out} is multiplied by $(A+1)$, where A is the gain of the auxiliary amplifier:

$$R_{out} = (A + 1)g_{m9}r_Ar_B \quad (1.34)$$

where r_A corresponds to $R_{S9,OL}$, and:

$$r_B = r_{bc}||r_{folded} = [(1/g_{m15}) + r_{o11}]||r_{folded} \quad (1.35)$$

with:

$$r_{folded} = g_{m9}r_{o9}(R_1||R_2) \quad (1.36)$$

Combining these equations gives:

$$A_{OL} = g_{m,comp}R_{out} =$$

$$A_{OL} = g_{m,comp}(A + 1)g_{m9}R_{S9,OL}[((1/g_{m15}) + r_{o11}) \parallel (g_{m9}r_{o9}(R_1||R_2))]$$

Thus, the closed-loop resistance of the initial relation can be calculated. This resistance has a very small value and, consequently, G_{S9} is very large. The conductance G_{D5} is:

$$G_{D5} = \frac{1}{r_{o7} + r_{o5} + g_{m5}r_{o5}r_{o7}} \quad (1.37)$$

which gives a very small value. Therefore, $G_{S9} >> G_{D5}$.

For the condition $G_{S3}r_{o1} >> 1$, G_{S3} is calculated using the small-signal model [1]:

$$G_S = \frac{is}{vs} = \frac{g_{s1} + 1/r_{o1}}{1 + Z_L/r_{o1}} \quad (1.38)$$

Substituting gives:

$$G_{S3} = \frac{g_{s3} + 1/r_{o3}}{1 + 1/(r_{o3}(G_{S9} + G_{D5}))} \approx \frac{g_{s3}}{1 + 1/(r_{o3}(G_{S9} + G_{D5}))} \approx g_{s3} \quad (1.39)$$

From simulations, the resistance seen at the drains of Q_3, Q_4, Q_5, and Q_6 is sufficiently larger than the resistance seen at the sources of Q_9 and Q_{10}. Therefore, G_m is close to g_{m1}, as a result, the output-branch current can be scaled down by approximately a factor of 25, without significantly degrading the circuit transconductance. In this way, the output transistors maintain very low power consumption. The main source of power consumption in the amplifier is therefore the input branches, equal to $P_{DISS} = 2I_{BIAS}V_{DD}$.

1.4.2 Noise Analysis

The noise of the cascode transistors contributes very little to the total amplifier noise [14], because the cascode transistors cancel their own noise-current sources. Therefore, the noise contribution comes from the non-cascode transistors, and the noise of Q_3 and Q_4 is neglected.

The channel thermal noise of a MOS transistor can be modeled as a noise-current source between the source and drain terminals and is given by:

$$i_n^2 = 4kT\gamma g_m \tag{1.40}$$

where k_B is Boltzmann's constant, T is temperature, γ is the noise coefficient, and g_m is transconductance.

Flicker noise can be modeled as a noise-current source between the source and drain terminals and is given by:

$$i_n^2 = gm^2 \frac{K}{C_{ox}WLf} \tag{1.41}$$

where K is a technology-dependent constant, C_{ox} is the oxide capacitance, W is width, L is length, and f is frequency.

Because the output current is approximately 25 times smaller than the input current, the noise contribution of the output transistors is very small and can be neglected. Transistors Q_7 and Q_8 operate as source-degeneration resistors, and the transconductance of the entire composite cascode topology (Q_5, Q_6, Q_7, and Q_8) is reduced. Consequently, their noise is reduced. Large lengths were used to further reduce the flicker noise of these transistors. Their widths were kept small in order to keep their transconductance low, as discussed earlier.

The coefficient γ in subthreshold operation is equal to $1/(2\kappa)$, where κ is a technology-dependent subthreshold coefficient. Above threshold, $\gamma \approx 2/3$. For simplicity, $\kappa \approx 0.75$ is assumed. With these simplifications, the total input-referred noise of the amplifier is:

$$V_n^2 \approx \frac{k_B T}{g_{m1}^2}\left(\frac{16 g_{m1}}{3}\right) + \frac{2K_p}{WLC_{oxp}f} \tag{1.42}$$

The coefficient K is smaller for PMOS transistors. Therefore, PMOS input transistors with large widths were selected to reduce flicker noise. Their lengths were kept small, taking area consumption and transconductance maximization into account. Thus, the input transistors operate in the subthreshold region, where the Inversion Coefficient $IC = \frac{I_D}{I_{D0}\frac{W}{L}}$ should be less than 0.1. Since the transistors operate in the subthreshold region, the current is approximately:

$$I_{D1} = I_{D0}\frac{W}{L}e^{\frac{V_{GS}-V_{TH}}{nV_T}}\left(1 - e^{-\frac{V_{DS}}{V_T}}\right) \approx I_{D0}\frac{W}{L}e^{\frac{V_{GS}-V_{TH}}{nV_T}} \tag{1.43}$$

where I_{D0} is a technology-dependent subthreshold coefficient. The transconductance is:

$$g_{m1} = \frac{I_{D1}}{nV_T} \tag{1.44}$$

The input-referred noise of the CCIA is calculated as follows [3]:

$$V_{CCIA}^2 = \left(\frac{C_{in} + C_{feedback} + C_p}{C_{in}}\right)^2 V_{OTA}^2 \tag{1.45}$$

where C_p is the input parasitic capacitance. Therefore, it is desirable to have $C_{in} >> C_{feedback}$ in order to reduce the noise contribution of the feedback system.

1.4.3 Common-Mode Rejection Analysis

For the common-mode rejection, power-supply rejection, and robustness specifications, the mismatch of matched transistors must be as small as possible. The way in which the dimensions of matched devices were selected is analyzed later in Section 1.5. More specifically, the dimensions and multipliers of the current mirror that biases the amplifier input are discussed. Similar reasoning was used for the other matched devices. It should be noted that the largest mismatch appears in the output cascode current mirror consisting of Q_{12}, Q_{13}, Q_{14}, and Q_{15}. This occurs because it operates closer to the subthreshold region than to saturation. As a result, current varies exponentially with threshold voltage, making differences between matched transistors more pronounced.

The common-mode gain due to the feedback-system capacitances is calculated according to Figure 1.8 [3]:

$$A_{cm} = \frac{C_{in2}}{C_{in2} + C_{f2}}\left(1 - \frac{C_{in1}}{C_{in2}}\frac{C_{f2}}{C_{f1}}\right) \tag{1.46}$$

If the capacitors are perfectly matched, then $C_{in1} = C_{in2}$ and $C_{f1} = C_{f2}$, making $A_{cm} = 0$ and the CMRR infinite. In practice, imperfections must be considered:

$$C_{in} = \frac{C_{in1} + C_{in2}}{2}, \quad \Delta C_{in} = C_{in1} - C_{in2} \tag{1.47}$$

$$C_f = \frac{C_{f1} + C_{f2}}{2}, \quad \Delta C_f = C_{f1} - C_{f2} \tag{1.48}$$

Because $\Delta C_f << C_f$ and because of the high differential gain of the amplifier, the common-mode gain can be approximated by:

$$A_{cm} \approx \frac{\Delta C_f}{C_f} - \frac{\Delta C_{in}}{C_{in}} \tag{1.49}$$

Since $C_{in} >> C_f$, it can be further approximated as:

$$A_{cm} \approx \frac{\Delta C_f}{C_f} \tag{1.50}$$

Therefore, increasing C_f can improve the CMRR, but this reduces the differential gain. To preserve the desired differential gain, C_{in} must be increased proportionally. Thus, there is a trade-off between CMRR and area consumption.

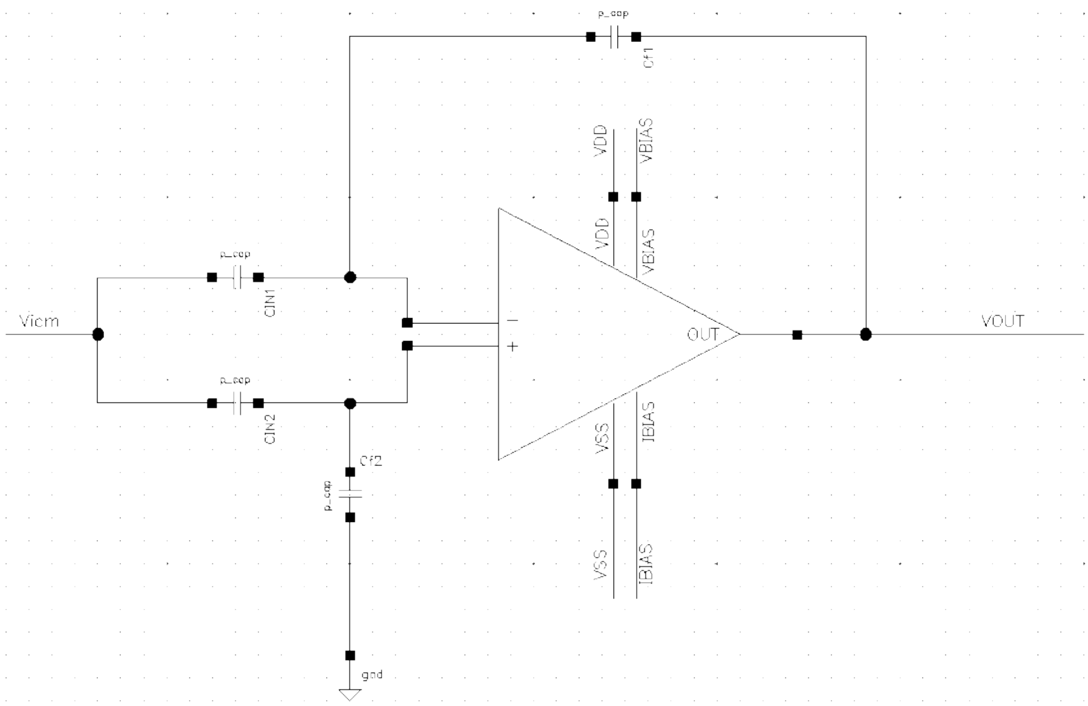


Figure 1.8: Common-mode gain due to capacitance mismatch

1.4.4 Gain, Bandwidth, and Phase-Margin Analysis

The amplifier gain is determined by the ratio C_1/C_3, as shown in Figure 1.7. As in previous amplifier designs for cochlear implants [15] [7], the gain in this work was selected slightly above 40 dB.

Including the high-pass output filter, the bandwidth is defined by the following two poles:

$$s = -\frac{C_3 G_m}{C_1 C_5}, \quad s = -\frac{1}{RC_6}$$

Therefore, the corresponding frequencies are:

$$f_1 = \frac{1}{2\pi R C_6}, \quad f_2 = \frac{C_3 G_m}{2\pi C_1 C_5}$$

where R is the value of the MOS pseudo-resistor and is selected around $4G\Omega$. In this way, the two main poles are placed at 20 Hz and 20 kHz, defining the amplifier bandwidth. This bandwidth corresponds to the audible frequency range of the human ear.

For the phase margin, it is very important that the gain drops with a slope of $-20\,\mathrm{dB/decade}$ before reaching the unity-gain frequency. This ensures that the phase margin remains high, preventing excessive oscillations and instability. Undesired poles and zeros due to parasitic elements were adjusted so that they appear after the unity-gain frequency. As discussed in Chapter 2, increasing C_5 reduces the unity-gain frequency and increases the phase margin, whereas a smaller capacitive load increases bandwidth but reduces phase margin. Considering all these factors, the phase margin was set to approximately 72°.

1.4.5 Layout Analysis

For the layout of the amplifier, the bias circuit, the feedback circuit, and the output filter, the following design steps were followed:

- Matched transistors were placed as close as possible to reduce process variations.
- Matched transistors in current mirrors were given the same orientation, source-to-drain, so that their current direction is the same.
- Metal layers from metal 1 to metal 3 were used, with only a few exceptions where metal 4 was required.
- Metal layers 1 and 3 were mainly used vertically, while metal 2 was used horizontally in order to reduce metal-corner routing.
- Guard rings were used for improved noise isolation, better matching, and more stable operation.
- The common-centroid technique was applied to the differential pair, the feedback and input capacitors, and wherever else necessary, to improve matching, reduce sensitivity to process variations, and optimize noise.
- Particular attention was given to the symmetric layout of matched-device connections.
- Poly-silicon interconnects were avoided in order to minimize ohmic losses.

Figures 1.9 and 1.10 show two examples from the present design where the above steps were applied. The complete physical design of the amplifier, bias circuit, feedback circuit, and output filter is analyzed in the next chapter.

Figure 1.9 shows the physical design of the input differential pair using the common-centroid technique. The metal surrounding the transistors is the n-guard ring, the blue color represents n-type diffusion. The upper layer of the guard ring is metal 1. Cyan corresponds to metal 3, yellow to metal 2, and gray to metal 1. Metal 1 was used horizontally only for short interconnections where no other metal-1 connection was expected to pass. Metal 3 was used for the long vertical interconnections outside the transistors. Green shows the polysilicon at the transistor gates, under which the gate oxide is located. The white background corresponds to the n-well, while red indicates p-type diffusion at the drain and source of each transistor.

As shown, transistors Q_1 and Q_2, each with a multiplier of 8, were placed symmetrically and diagonally. Each of Q_1 and Q_2 has four transistors in the upper part and four in the lower part. Considering x and y axes that split Figure 1.9 into upper/lower and right/left regions, the variations cancel as follows:

$$Q_{1,left,variations} : 4(+\Delta y - \Delta x) \quad Q_{1,right,variations} : 4(-\Delta y + \Delta x)$$

Therefore:

$$Q_{1,total,variations} : 4(+\Delta y - \Delta x) + 4(-\Delta y + \Delta x) = 0$$

The variations for the transistors composing Q_2 are similarly calculated to be zero.

Figure 1.10 shows the output current mirror consisting of transistors Q_{14} and Q_{15}. As before, the white background corresponds to the n-well, while the dotted background is the p-type substrate. The distances between the transistors were selected to be the minimum values allowed by the technology. The transistors were placed with the same orientation, source to drain. Q_{14} consists of the two inner transistors, while Q_{15} consists of the two outer transistors. Considering only the y-axis in this case, the variations cancel.

1.5 Bias Circuit

This chapter section analyzes the designed bias circuit. The bias circuit provides the amplifier bias current. It also generates the bias voltage V_{BIAS} for transistors Q_{17} and Q_{19}, as well as the output high-pass filter bias voltages V_{BIAS1} and V_{BIAS2} for M_5 and M_6, respectively.

The circuit consists of two intentionally mismatched transistors, Q_{25} and Q_{26}, with Q_{26} usually having approximately four times the width of Q_{25}. A resistor R is connected in series with the source of Q_{26}. As shown below, R determines the bias current I_{BIAS}. These two transistors and the resistor form a Widlar current source. In addition, there

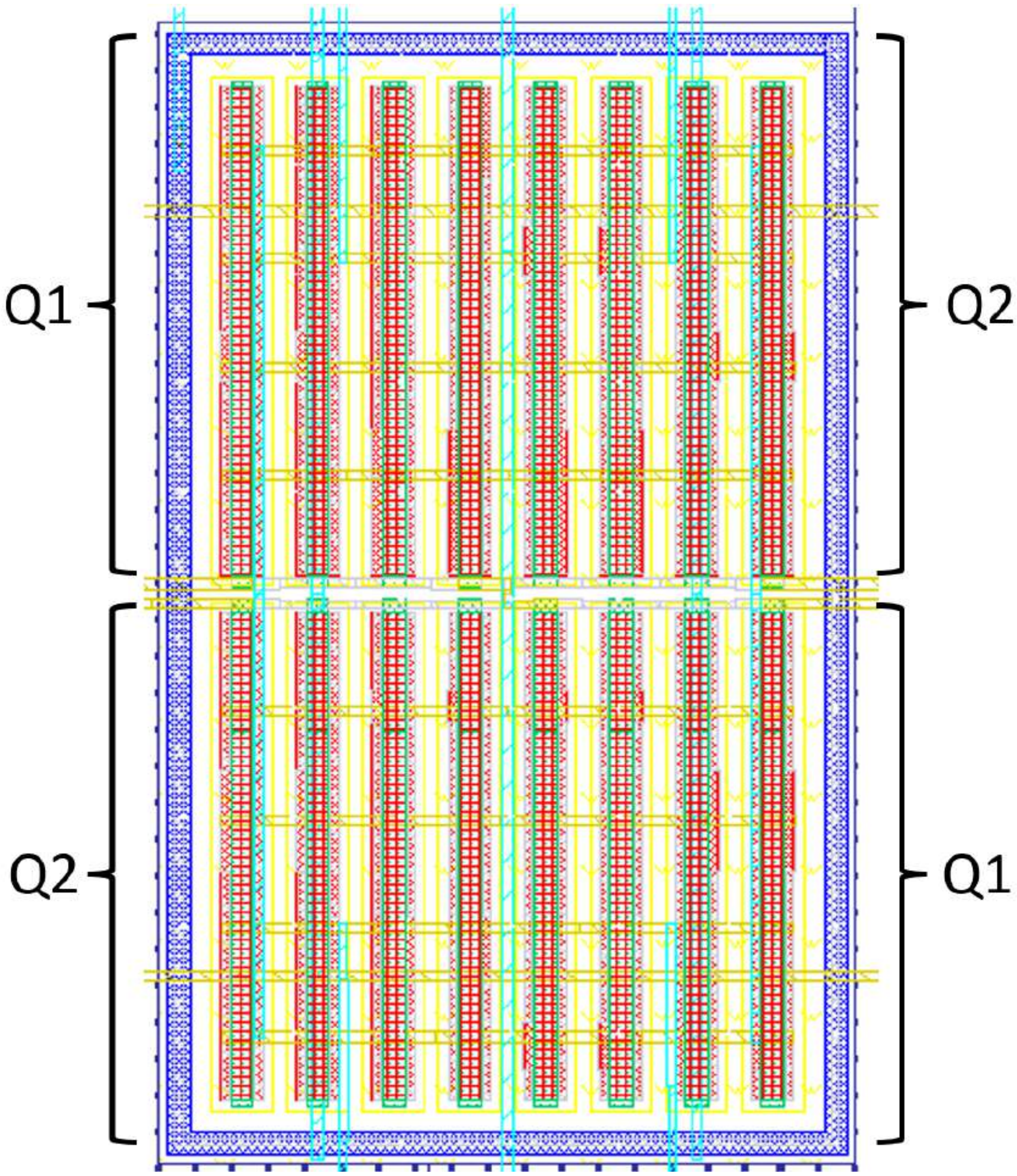


Figure 1.9: Physical design of the input differential pair

are two cascode transistors, Q_{23} and Q_{24}. They are used to reduce the body effect in Q_{26}, and Q_{23} provides a bias voltage to Q_{25}. Finally, a PMOS current mirror biases Q_{25} and Q_{26} with current I_{BIAS} and provides a bias line for the amplifier.

The objective is to obtain the bias voltages $V_{BIAS1} = 410mV$, $V_{BIAS2} = 1.37V$, and $V_{BIAS} = 1V$, together with a bias current $I_{BIAS} = 400nA$. Since $V_{GS26} < V_{GS25} < V_{th}$, Q_{25} and Q_{26} are intended to operate in the subthreshold region, so:

$$I_{DS26} = I_{DS25} = I_o e^{\frac{V_{GS}-V_{th}}{nV_T}} \left(1 - e^{\frac{V_{DS4}}{V_T}}\right) \tag{1.51}$$

$$I_o = \mu C_{ox} \frac{W}{L} V_T^2 e^{1.8} \tag{1.52}$$

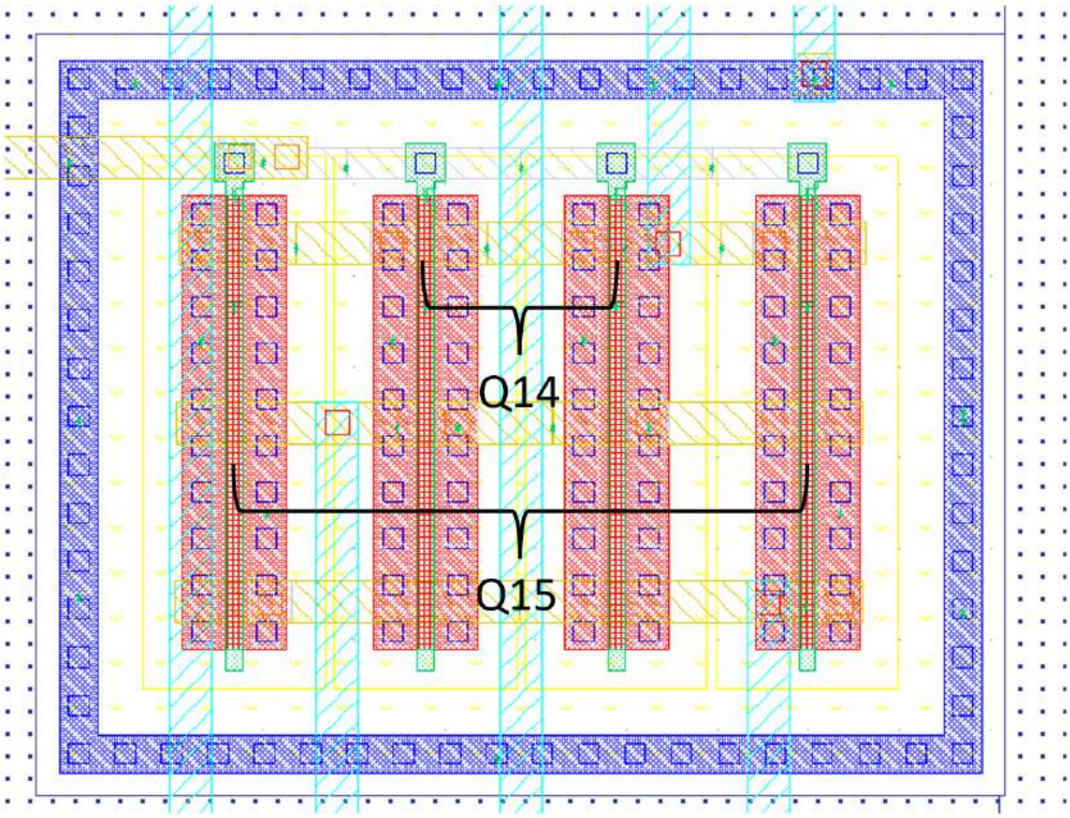


Figure 1.10: Physical design of the output current mirror

Because $\left(1 - e^{V_{DS4}/V_T}\right) \approx 1$:

$$I_{DS25} = I_o e^{\frac{V_{GS} - V_{th}}{nV_T}} \tag{1.53}$$

Substituting the desired values and selecting $n = 1.5$, the dimensions of Q_{25} are obtained as $W/L \approx 125$.

Furthermore:

$$V_{GS26} + I_{BIAS}R = V_{GS25} =>$$

$$V_{th26} + nV_T \ln\left(\frac{I_{BIAS}}{I_{o26}}\right) + I_{BIAS}R = V_{th25} + nV_T \ln\left(\frac{I_{BIAS}}{I_{o25}}\right) =>$$

$$R = \frac{nV_T}{I_{BIAS}} \ln\left(\frac{I_{o26}}{I_{o25}}\right) + \frac{V_{th25} - V_{th26}}{I_{BIAS}}$$

Selecting four-times larger dimensions for the corresponding device gives:

$$\frac{I_{o4}}{I_{o3}} = 4$$

Therefore:

$$R = \frac{nV_T}{I_{BIAS}} \ln(4) + \frac{V_{th25} - V_{th26}}{I_{BIAS}} \tag{1.54}$$

The term $(V_{th25} - V_{th26})/I_{BIAS}$ gives a fairly large negative value, which reduces the resistance by approximately $35k\Omega$. The widths and lengths of the remaining transistors, which operate in saturation, were calculated in a similar manner. Then, based on simulations, their dimensions were slightly tuned so that the desired specifications could be achieved. The final dimensions and operating regions are shown in Table 1.3. The selected resistor value was $97.5005k\Omega$. It is integrated on chip and implemented using n-well diffusion.

Table 1.3: Bias-circuit transistor dimensions

Transistor	W/L	Operating region
Q_{20}, Q_{22}	$(5/10) \times 2$	saturation
Q_{21}	$(5/10) \times 4$	saturation
Q_{23}, Q_{24}	$(0.5/8.76)$	saturation
Q_{25}	$(5.485/0.18) \times 4$	subthreshold
Q_{26}	$(23.05/0.18) \times 4$	subthreshold

Figure 1.12 shows the input differential pair together with the current mirror that biases it. Transistor Q_{22}, which is part of the bias circuit, must have aspect ratio $(W/L) = 1$ in order to generate the bias voltage V_{BIAS}.

In the circuit of Figure 1.12, assuming that the bias current through Q_{20} is equal to I_{BIAS}, the desired current for each transistor of the differential pair is achieved because the bias transistor Q_{21} has twice the W/L ratio. This current is equal to I_{BIAS}. An alternative would be $(W/L)_{Q20} = (W/L)_{Q21} = 1$ with the bias circuit operating at $2I_{BIAS}$, but this would increase power consumption and is therefore undesirable. Possible mismatches that can degrade current-mirror matching [16], as well as the reason for selecting the specific dimensions, are analyzed below. Since the mirror transistors operate in saturation:

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{th})^2 = \frac{\beta}{2}(V_{GS} - V_{th})^2 \tag{1.55}$$

If $y = f(x_1, x_2, \ldots, x_n)$ is a function of n independent variables, the variance is:

$$\sigma_y^2 = \sum_{m=1}^{n}\left(\frac{dy}{dx_m}\right)^2 \sigma_{x_m}^2 \tag{1.56}$$

Substituting $n = 2$, $x_1 = \beta$, and $x_2 = V_{th}$ gives:

$$\sigma_{I_D}^2 = \left(\frac{(V_{GS} - V_{th})^2}{2}\right)^2 + \left(\frac{\beta}{2}\right)^2 (-2(V_{GS} - V_{th}))^2 \sigma_{V_{th}}^2 \tag{1.57}$$

Dividing by I_D^2 gives:

$$\left(\frac{\sigma_{I_D}}{I_D}\right)^2 = \left(\frac{\sigma_\beta}{\beta}\right)^2 + \left(\frac{2}{V_{GS} - V_{th}}\right)^2 \sigma_{V_{th}}^2 = \sigma_r^2 \tag{1.58}$$

where:

$$\left(\frac{\sigma_\beta}{\beta}\right)^2 = \frac{A_\beta^2}{WL} \propto \frac{1}{WL}, \quad \sigma_{V_{th}}^2 = \frac{A_{Vth}^2}{WL} \propto \frac{1}{WL} \tag{1.59}$$

Therefore, increasing the width and length of a transistor reduces its mismatch. It is also important to use the same width and length for transistors in the same current mirror. Otherwise, the gain-factor mismatch coefficient (A_β) and the threshold-voltage mismatch coefficient (A_{Vth}) change and are no longer identical for the mirrored transistors, since width and length affect V_{th} and β. If the objective is to generate a current that is M times larger, it is preferable to use the corresponding multiplier.

If M and N are the multipliers of Q_{20} and Q_{21}, respectively, the output-current mismatch of the current mirror, namely the current entering the differential pair, is:

$$\sigma^2_{Iout} = \sigma^2_{rQ20} + \sigma^2_{rQ21}, \quad \sigma^2_{rQ20} = \frac{\sigma^2_r}{M}, \quad \sigma^2_{rQ21} = \frac{\sigma^2_r}{N} \tag{1.60}$$

Thus:

$$\sigma_{Iout} = \sigma_r \sqrt{\frac{M+N}{MN}} \tag{1.61}$$

Therefore, provided both transistors have the same width and length and therefore the same relative mismatch σ_r, the output-current mismatch depends on their multipliers.

Considering all the above, as well as layout matching, the same width and length were selected for Q_{20} and Q_{21}. The multiplier of Q_{20} was set to 2 and the multiplier of Q_{21} to 4. Simulations showed a current mismatch of 0.075%, which is extremely low.

1.6 Reference Voltage Circuit

The reference-voltage circuit of the present work is used to provide a stable reference voltage to the amplifier input, while maintaining high power-supply rejection and low input noise. The circuit uses the same topology as the circuit presented in [17]. Conventional reference circuits usually attempt to combine a parameter that is proportional to absolute temperature (PTAT) with another that is complementary to absolute temperature (CTAT). More specifically, CTAT (Complementary-to-Absolute-Temperature) and PTAT (Proportional-to-Absolute-Temperature) refer to voltages or currents with opposite temperature dependence that are combined to generate a temperature-independent reference voltage. Thus, $V_{REF} = CTAT\ Component + PTAT\ Component$, where the CTAT component decreases with temperature and the PTAT component increases with temperature.

However, this usually leads to increased complexity and power consumption.

Figure 1.13 presents a voltage-reference circuit based on a weighted difference of V_{GS} voltages, resulting in lower power consumption and improved power-supply rejection (PSRR). The circuit operates in the subthreshold region to further reduce power consumption. It consists of start-up, bias circuit, and core sections. The start-up circuit consists of transistors Q_{27}, Q_{28}, and Q_{29}. It prevents the circuit from settling in a zero-current state during startup by activating the bias circuit. The bias circuit consists of transistors Q_{30}-Q_{33}. The bias circuit generates a subthreshold bias current $I_{sub} = \frac{V_{GS31}-V_{GS30}}{R_1}$, which is mirrored to the core through Q_{37}. It ensures stable operation by providing temperature-independent biasing to the core transistors. Finally, the core consists of transistors Q_{34}–Q_{37} and generates the reference voltage V_{REF} using the weighted difference of the V_{GS} voltages of MOS transistors operating in the subthreshold region. The reference voltage is calculated as:

$$V_{REF} = V_{GS34} + V_{GS35} - |V_{GS36}|$$

In the present work, the desired power-supply rejection of the reference-voltage circuit, so that it does not degrade the PSRR, must be -70 dB over the 20 Hz–20 kHz operating range. As shown in the next chapter, the PSRR of the overall system reaches 70 dB. Because V_{REF} biases the amplifier input, any unwanted supply signal is injected into the amplifier input and amplified by 40 dB. Therefore, if the reference-voltage circuit rejects the supply by -70 dB, this signal is amplified by 40 dB, keeping the amplifier supply rejection at -30 dB and therefore the PSRR at 70 dB. The calculation of the PSRR of the circuit in Figure 1.13 is presented in [17] and is not repeated here. The temperature variation is analyzed next.

The current generated by the bias circuit is:

$$I_{sub} = \frac{V_{GS31} - V_{GS30}}{R_1} \tag{1.62}$$

Using the subthreshold current expression:

$$I_D = \mu_n C_{ox} \frac{W}{L} V_T^2 \exp\left(\frac{V_{GS} - V_{th}}{nV_T}\right)$$

Solving for V_{GS} gives:

$$V_{GS} = nV_T \ln\left(\frac{I_D}{\mu_n C_{ox} \frac{W}{L} V_T^2}\right) + V_{th} \tag{1.63}$$

Performing the calculations for the current produced by the bias circuit leads to:

$$\frac{nkT_0}{qR_1} \ln\left(\frac{(W/L)_{32}}{(W/L)_{31}}\right) \frac{T}{T_0} = I_{sub}(T_0)\frac{T}{T_0} \tag{1.64}$$

Taking the derivative of V_{REF} with respect to temperature and using the above expressions gives:

$$\frac{dV_{REF}}{dT} = \frac{dV_{GS34}}{dT} + \frac{dV_{GS35}}{dT} - \frac{d|V_{GS36}|}{dT} => \tag{1.65}$$

$$\frac{dV_{REF}}{dT} = \frac{nk}{q}\ln\left(\frac{MI_{sub}(T_0)\left(\frac{W}{L}\right)_{36}}{\mu_n(T_0)(T_0)^{\beta_{\mu_n}+1}C_{ox}\left(\frac{k}{q}\right)^2\left(\frac{W}{L}\right)_{35}\left(\frac{W}{L}\right)_{34}}\right) + nk\frac{q}{T}(\beta_{\mu_n}-1) - \beta_{Vthn}$$

Setting this equal to zero gives:

$$\ln\left[\frac{\left(\frac{W}{L}\right)_{36}}{\left(\frac{W}{L}\right)_{34}\left(\frac{W}{L}\right)_{35}}\right] = \ln\left[\frac{\mu_n(T_0)T_0^{\beta_{\mu_n}+1}C_{ox}(k/q)^2}{MI_{sub}(T_0)T_0^{\beta_{\mu_n}-1}}\right] - \beta_{\mu_n} + 1 + \frac{q}{nk}(\beta_{Vthn}) \tag{1.66}$$

where:

$$\mu_n(T) = \mu_n(T_0)\frac{T}{T_0}^{-\beta_{\mu_n}}$$

$$V_{thn}(T) = V_{thn}(T_0) - \beta_{vthn}(T - T_0)$$

- $\mu_n(T_0) \approx 350cm^2/sV$: electron mobility at reference temperature T_0.
- T_0: reference temperature.
- $\beta_{\mu_n} \approx 1.5$: electron-mobility exponent.
- $C_{ox} = 9.32fF/\mu m^2$: oxide capacitance.
- $k = 1.3810^{-23}$: Boltzmann constant.
- $M = 0.25$: bias-current scaling coefficient.
- $I_{sub}(T_0) = 520nA$: subthreshold bias current at reference temperature T_0.
- $\beta_{Vthn} \approx -1mV/K$: temperature coefficient of the threshold voltage.

Based on the above, the first part of the equation should be approximately equal to 7.8. However, because this equation is not perfectly accurate, the dimensions were further tuned through simulations, resulting in a value of approximately 6.8 for the first part of the equation. The selected dimensions are shown in Table 1.4. In addition to the small temperature variation, the output-transistor dimensions were selected while also targeting a reference voltage of $900mV$. Capacitor C_7 at the output has a value of $125nF$ and is off-chip, which is not ideal because of its size [18]. It was used to form, together with the output resistance, a low-pass filter that removes input noise and further improves power-supply rejection. Without the capacitor, the circuit achieves power-supply rejection of $-44dB$ in the desired operating range. The pole was set on the order of mHz, and with a $20dB$/decade roll-off, the desired power-supply rejection is achieved at $20Hz$ and the noise is filtered out in the operating frequency spectrum. Resistor R_1 has a value of $94.4k\Omega$ and is on-chip.

Table 1.4: Reference-voltage circuit transistor dimensions

Transistor	W/L
Q_{27}	(20/1)
Q_{28}	(3/1)
Q_{29}	(0.5/20)
Q_{30}	$(20.1/1) \times 3$
Q_{31}	(19.8/1)
Q_{32}, Q_{33}	$(2.5/1) \times 8$
Q_{34}	(1/6)
Q_{35}	(1/9.95)
Q_{36}	(15/1)
Q_{37}	$(2.5/1) \times 2$

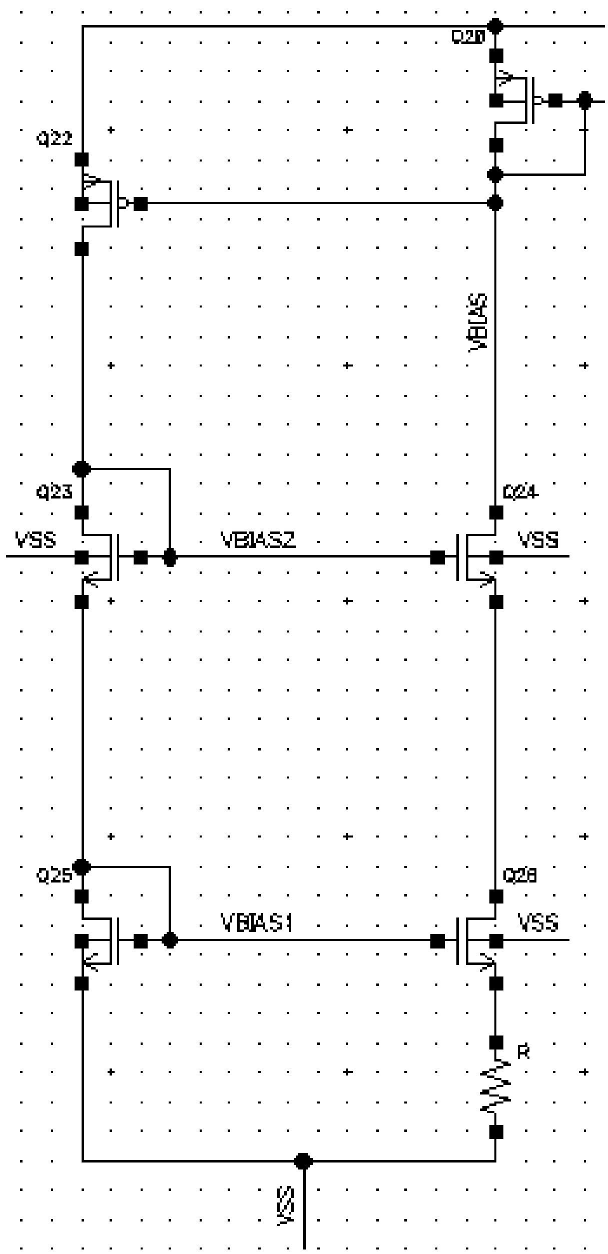


Figure 1.11: Bias circuit

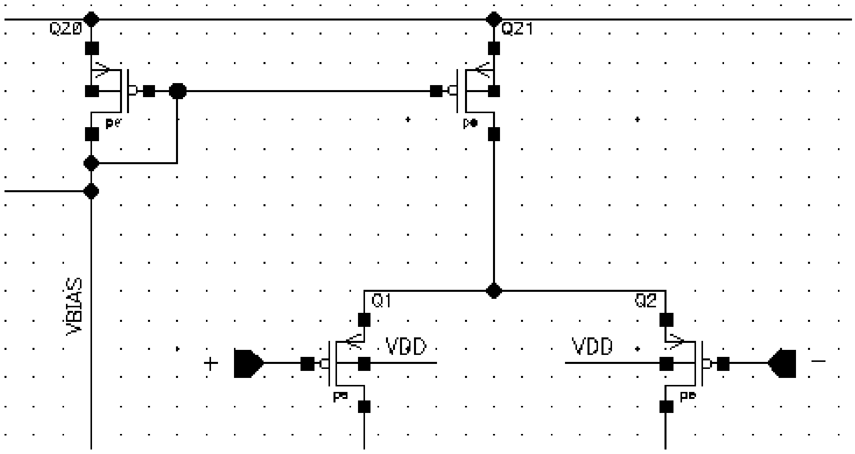


Figure 1.12: Current mirror of the input differential pair

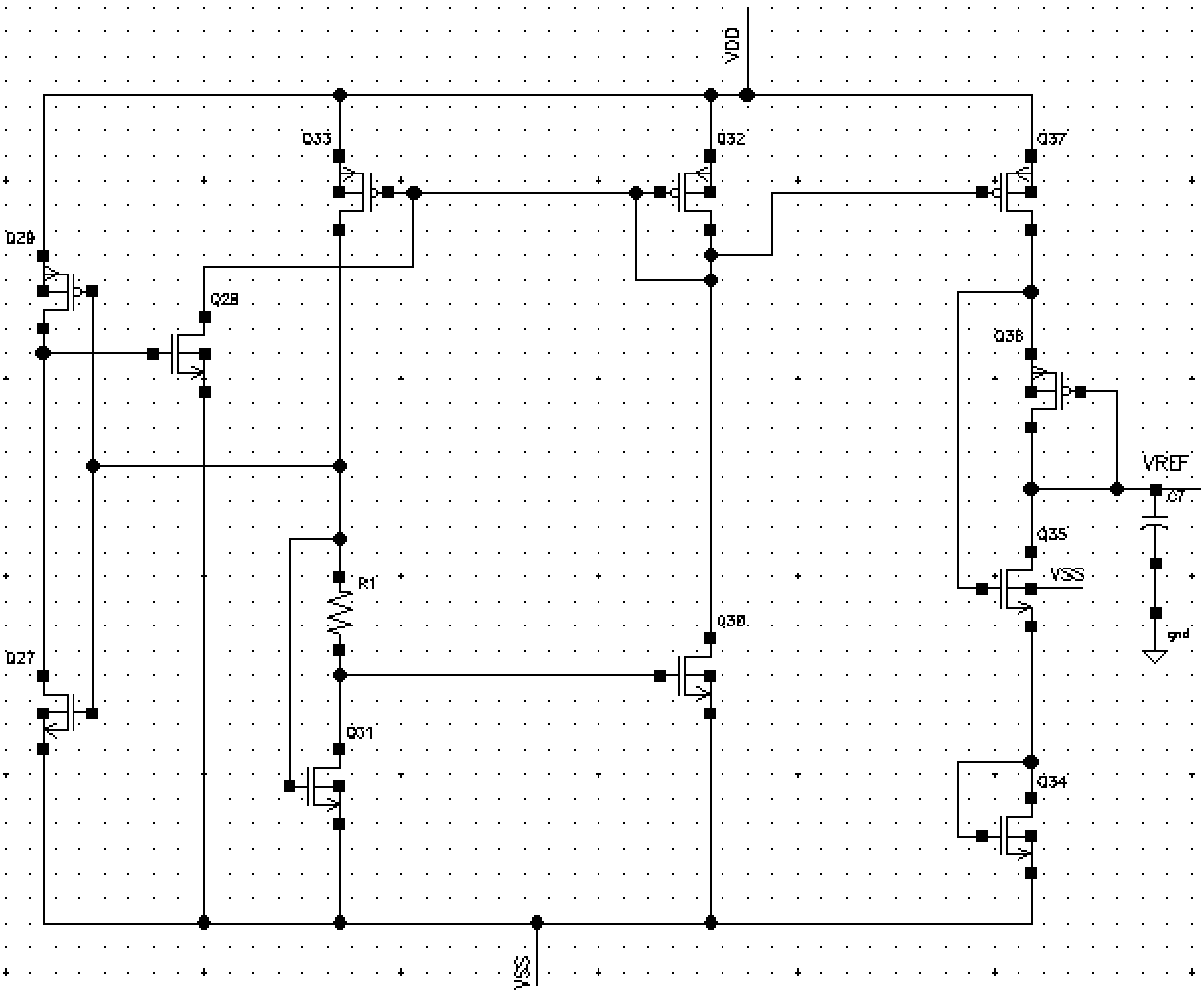


Figure 1.13: Reference voltage circuit

Chapter 2

Simulation Results

This chapter examines the operation of the designed analog front-end through simulations for all the different specifications that must be satisfied. First, the schematic-level results are analyzed, and then they are compared with the corresponding post-layout results. The designed analog front-end was simulated in X-FAB CMOS $0.18\mu m$ technology. The measurement methodology is presented in more detail in Appendix A.

2.1 Power Consumption

2.1.1 Amplifier Power Consumption

The static power consumption of the amplifier alone, obtained through simulations, was equal to $1.539\mu W$. This value results from the sum of the currents of the two input branches, the two output branches, and the auxiliary amplifiers. As mentioned in the previous chapter, these values are approximately $2 \times 400nA$, $2 \times 16nA$, $2 \times 12nA$, respectively. Multiplying this total current by the $1.8V$ supply voltage gives the static power consumption. The maximum dynamic power consumption for the maximum amplifier input deviates only slightly from the static value, keeping the amplifier power consumption at the desired level.

2.1.2 Bias Circuit Power Consumption

According to simulations, the bias circuit consumes $1.44054\mu W$. This follows from the fact that the bias circuit consists of two branches, each carrying a current equal to $400nA$ in order to generate the desired bias current. Therefore, since the supply voltage is $1.8V$, the static power consumption is approximately $1.44\mu W$.

2.1.3 Reference Voltage Circuit Power Consumption

The power consumption of the reference voltage circuit was found through simulations to be $2.4588\mu W$, making it the least power-efficient circuit among the three. The reference voltage circuit consists of four branches. The start-up circuit carries a current of approximately $195nA$, the two branches of the bias circuit carry $520nA$ each, and the output branch carries $130nA$. Since the supply voltage is the same, the consumption is approximately $2.457\mu W$, a value that agrees with the simulation result.

2.1.4 Total Power Consumption

In conclusion, the total consumption of the designed analog front-end is calculated as $1.539\mu + 1.44054\mu + 2.4588\mu \approx 5.44\mu W$. In the amplifier design, as mentioned in the previous chapter, the output current is 1/25 of the input current, greatly reducing the power consumption. More specifically, in a conventional folded cascode amplifier, the output current would be approximately $2 \times 100nA$, increasing the output-stage power consumption from $0.057\mu W$ to $0.36\mu W$. In the bias circuit, the current could be reduced further, at the cost of a larger resistance, which would not be fully acceptable from a physical-design perspective. A second alternative would be to use a larger off-chip resistor, which is not preferred. The internal biasing of the amplifier's cascode current mirror eliminates the need for a complex bias-circuit design, which would require more branches to provide the desired bias voltages and therefore greater power consumption. Although the reference voltage circuit has the highest power consumption, it remains within the desired range. Its output current is 1/4 of the current mirror of the bias stage, thereby reducing the total consumption. The consumption could be reduced further by increasing the resistance value, but for the reasons already discussed this is not preferred. Another way to minimize consumption is to reduce the difference $V_{GS31} - V_{GS30}$. Through simulations, in order to avoid large asymmetries between the two branches and to keep the transistor dimensions within reasonable values, a difference of approximately $50mV$ was selected.

2.2 Preamplifier Circuit–Bias Circuit

For the simulations in this section, the preamplifier and the bias circuit were used together with a DC voltage source that generates the bias voltage at the non-inverting input of the amplifier. The simulations including the reference voltage circuit are analyzed in the last section of the chapter, since that circuit negatively affects some specifications and, in its current form, cannot operate across different corners. Therefore, simulations were carried out using a DC voltage source in order to demonstrate the amplifier capabilities for an optimized reference voltage circuit design and to provide a clearer comparison with works that focus exclusively on amplifier design.

2.2.1 Input-Referred Noise of the Amplifier

Input-referred noise is the total noise generated by the amplifier referred to its input. It allows the degradation of the input signal due to noise to be evaluated, which is critical in applications such as biomedical devices where the signals are extremely weak. As discussed in previous Chapter, the main sources of this noise are thermal noise and flicker noise. To reduce noise, techniques such as current scaling, composite cascode, large-aspect-ratio PMOS input transistors, maximization of amplifier transconductance, and capacitive feedback were used.

Figure 2.1 shows the amplifier input-referred noise. At low frequencies, the increase is due to flicker noise, which increases as frequency decreases, whereas thermal noise is independent of frequency. Noise in an amplifier is often analyzed in terms of power spectral density, which describes how much noise power exists per unit bandwidth. The rms value of the noise over the frequency interval $f_1 - f_2$ is given by:

$$V_{n,rms}(f_1 - f_2) = \sqrt{\int SD^2 \, df}$$

- SD: Noise spectral density ($V/\sqrt{Hz}$)
- f: Frequency

As shown in Figure 2.1, the noise at $20Hz$, $1kHz$, and $9kHz$ is approximately $482nV/\sqrt{Hz}$, $64nV/\sqrt{Hz}$, and $44nV/\sqrt{Hz}$, respectively.

An index for the balance between noise, current consumption, and bandwidth is the NEF (Noise Efficiency Factor). The index is given by:

$$NEF = V_{n,rms}\sqrt{\frac{2I_{tot}}{\pi U_T 4kTBW}}$$

where I_{tot} is the total supply current and BW is the amplifier bandwidth.

The smaller this value is, the more optimal the balance is. The NEF was calculated to be approximately 1.86, while the rms noise was $V_{n,rms} = 7.41\mu V$ over the $20Hz - 20kHz$ bandwidth. The relatively small value of the NEF is due to the large bandwidth and to the fact that a large portion of the flicker noise, up to $20Hz$, is not included.

2.2.2 Gain–Bandwidth

The gain and bandwidth of the amplifier without and with the high-pass output filter are shown in Figures 2.2 and 2.3, respectively. The test bench used for this measurement

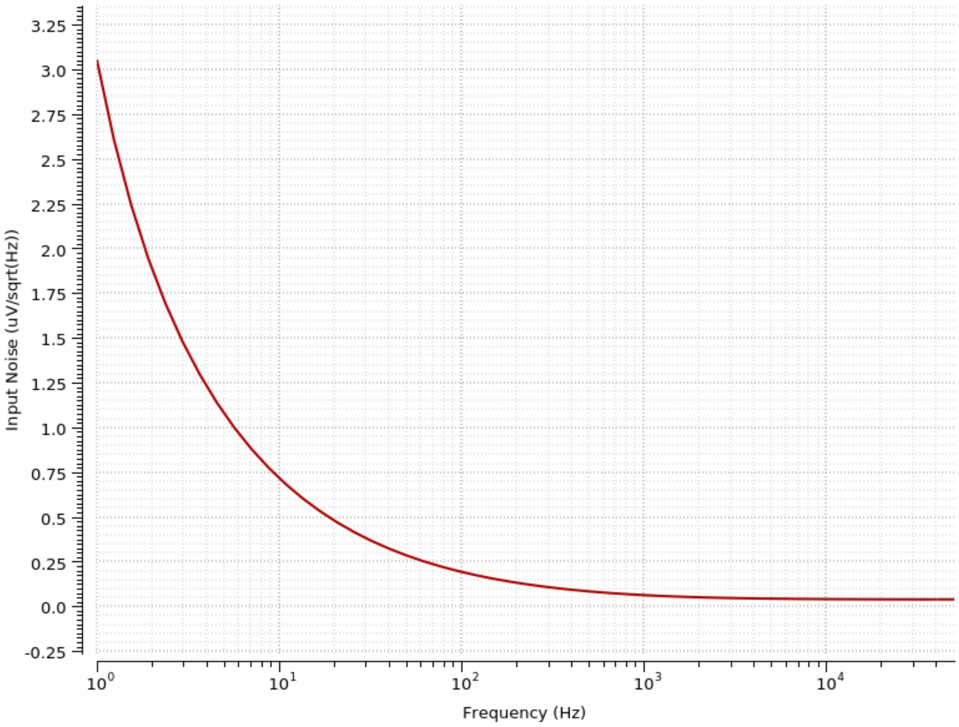


Figure 2.1: Input-referred noise spectrum density

is shown in Figure 1.7, with an AC voltage source connected to the INPUT terminal, the output of the reference voltage circuit connected to the V_{REF} terminal, and the bias circuit connected to the I_{BIAS} terminal, excluding transistor Q_{20}, which is included in the amplifier symbol, as shown in Figure 1.1.

As discussed in Chapter 1, the gain (C_1/C_3) was set slightly above $40dB$, amplifying the signal to the desired level for further processing by the sound processing unit. The bandwidth of the amplifier with the output filter is approximately from $20Hz$ to $20kHz$, covering the range of human hearing. Up to $2Hz$, where the first pole is located, the gain has a slope of $40db/$decade due to the two zeros at the origin. Then, from $2Hz$ to approximately $20Hz$, the slope is $20db/$decade.

It should be noted that if the high-pass filter at the output is removed and the first and second poles are shifted to lower frequencies, as shown in Figure 2.2, the amplifier can also be used in other biomedical applications. Indeed, as described in Chapter 1, to reduce the frequency of the second pole, the value of capacitor C_5 must be increased, without negatively affecting the phase margin. Thus, adapting the amplifier to similar biomedical applications is fairly straightforward.

2.2.3 Phase Margin

The phase margin of the amplifier is shown in Figure 2.4. The phase margin is the amount of phase remaining before the unity-gain frequency and is measured in an open-

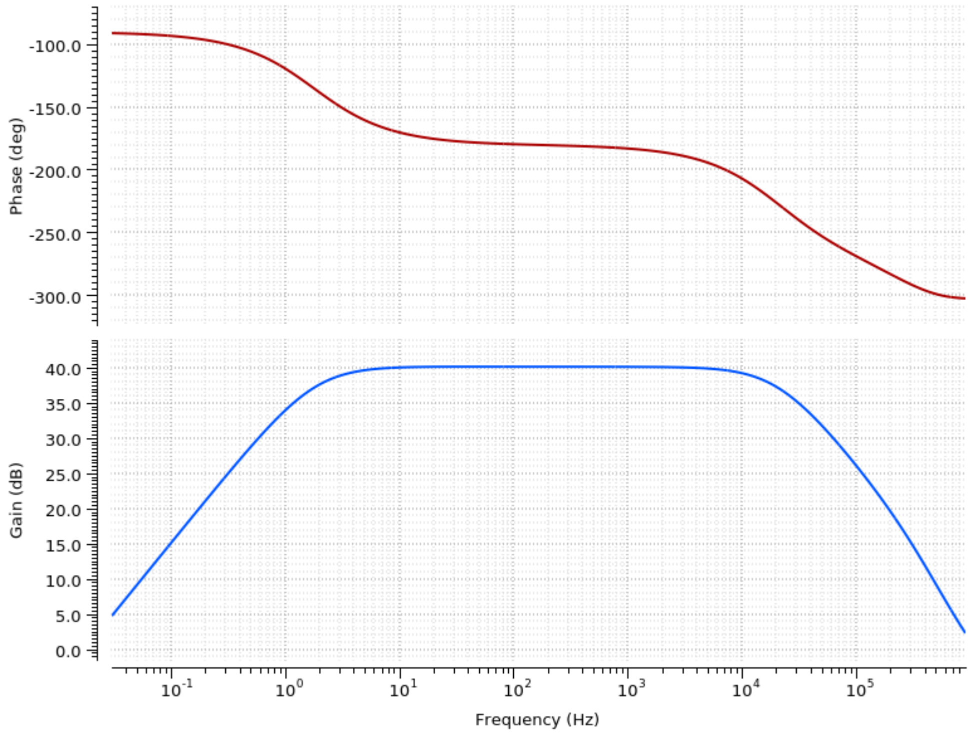


Figure 2.2: Frequency response without the output filter and phase diagram

loop system. It predicts whether the closed-loop system will be stable. A positive phase margin indicates that the system is stable, whereas a negative phase margin indicates instability. The phase margin should be greater than 45° to ensure a high level of stability. Its value can be calculated by finding the phase at the point where the gain becomes equal to zero for an inverting input. The phase margin of the amplifier was measured to be 72°. In addition, the amplifier gain margin was calculated to be approximately $12dB$, which is greater than $6dB$ and is therefore an acceptable value for stability.

The amplifier of the closed-loop system is configured in an inverting topology, and therefore, the phase is approximately $-180°$ in the mid-band frequency range. Moreover, due to the zero at $0Hz$ and the negative sign in the transfer function, the phase plot starts from $90° - 180° = -90°$, as shown in Figure 2.2.

2.2.4 Common-Mode Rejection Ratio

The Common-Mode Rejection Ratio (CMRR) expresses the ability of a differential amplifier to reject common unwanted signals while amplifying the differential signal. It is defined as:

$$CMRR = \frac{A_d}{A_{cm}} = 20 \log \left(\frac{A_d}{A_{cm}} \right) dB$$

where:

- A_d: Differential gain

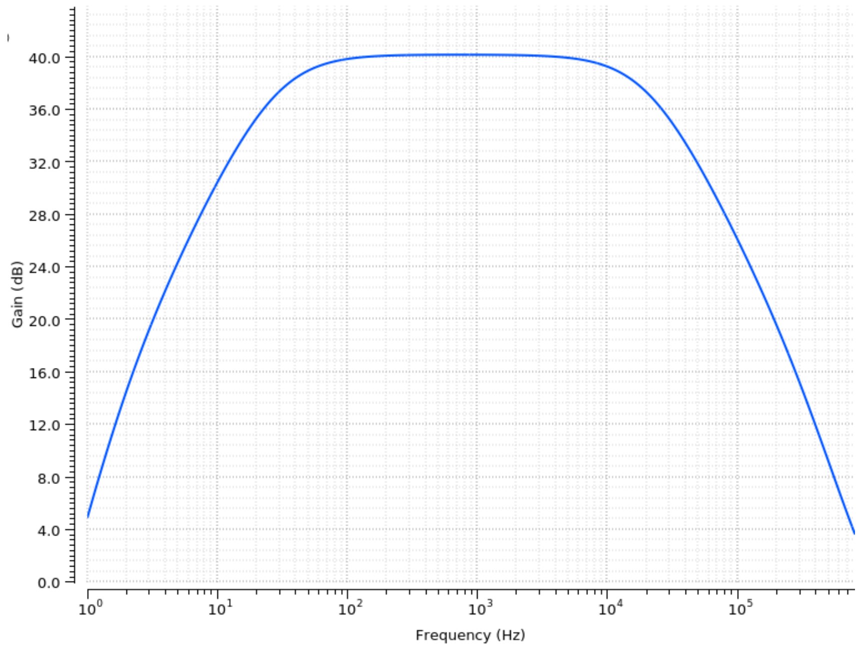


Figure 2.3: Frequency response of the amplifier with the output filter

- A_{cm}: Common-mode gain

The higher the CMRR, the better the rejection of unwanted noise. The CMRR is mainly determined by the matching of the input transistors. If the input bias current source is ideal, the common-mode gain is zero and the CMRR is infinite. In practice, however, the current source has finite output resistance and the common-mode gain is not zero. It decreases as this resistance increases and depends on the degree of asymmetry of the input transistors. Therefore, as analyzed in Chapter 1, the input transistors should be matched as closely as possible, also at the physical-design level. In addition, the length of the transistors that provide the input bias current was increased in order to increase their output resistance, and asymmetries in the capacitors of the feedback system were limited.

The CMRR is shown in Figure 2.5. For the test bench used to measure the common-mode gain, the two amplifier inputs were connected so that the input signal was common to both. The common-mode gain of the amplifier was then measured; since it is less than zero, it was subtracted, in dB, from the differential gain. The CMRR in the amplifier operating range achieves values from $117dB$ at $20Hz$ to $62dB$ at $20kHz$. In the critical operating range, the values are from $93dB$ at $500Hz$ to $67dB$ at $9kHz$, providing high common-mode rejection, especially at low frequencies.

2.2.5 Power-Supply Rejection Ratio

The Power-Supply Rejection Ratio (PSRR) is a measure of the amplifier's ability to suppress supply noise so that it does not affect the output. It is mathematically defined

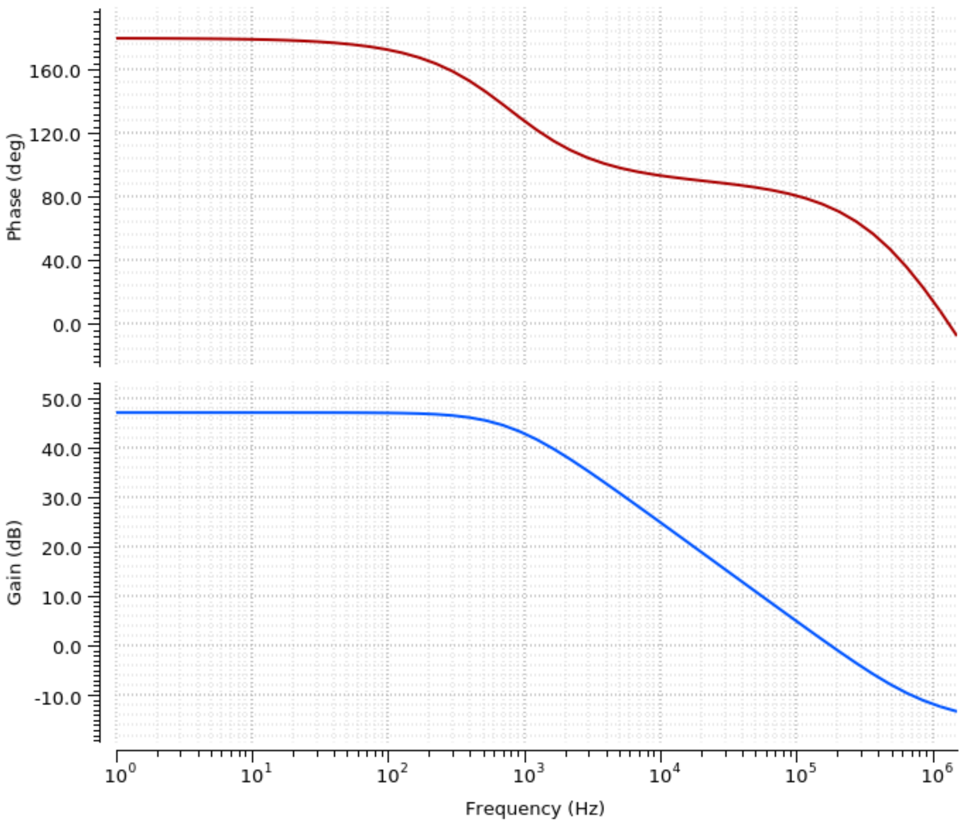


Figure 2.4: Phase margin

as:

$$PSRR = 20 \log \left(\frac{\Delta V_{supply} A_d}{\Delta V_{out}} \right)$$

A high PSRR indicates better rejection of unwanted supply signals, ensuring stable amplifier operation. Figure 2.6 shows the simulated PSRR. To calculate the gain from the supply, the test bench was configured accordingly, with an AC source connected in series with the supply, while the amplifier input was grounded. The gain from the supply to the amplifier output was then measured. Subtracting the supply gain ($-30dB$) from the differential gain ($40dB$) gives the PSRR. As shown, it remains stable, slightly above $70dB$, throughout the desired frequency range, which is a positive result.

One reason for the good PSRR response is that the folded cascode topology provides high rejection of signals from the supply. In addition, its tuning was performed while considering that device matching, high gain, and high bias current play an important role in increasing it. Finally, as mentioned in previous Chapter, the bias and reference voltage circuits were designed so that they do not degrade the PSRR achieved by the amplifier.

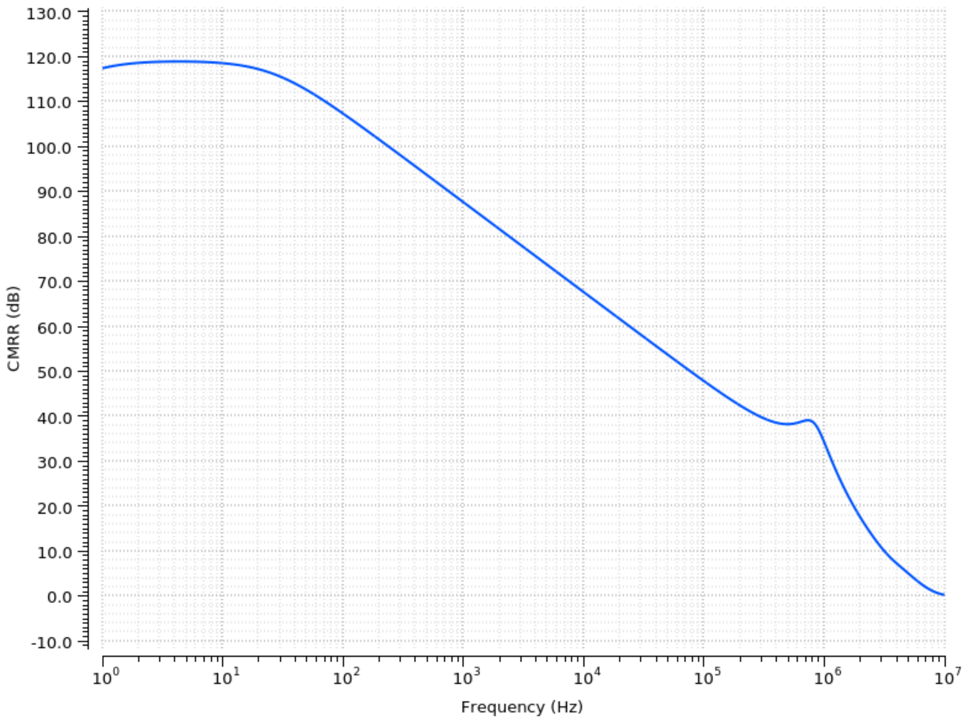


Figure 2.5: Common-mode rejection ratio

2.2.6 Harmonic Distortion

Total harmonic distortion (THD) is the measure of harmonic distortion present in a signal and is defined as the ratio of the sum of the powers of all harmonic components to the power of the fundamental frequency. More specifically, when a sinusoidal signal of frequency f passes through a non-ideal, nonlinear device, nf harmonics of the original frequency are added. Total harmonic distortion is a measure of this additional content that does not exist in the input signal. The measurement is usually defined as the ratio of the RMS amplitude of a set of higher-frequency harmonics to the RMS amplitude of the fundamental frequency. It is mathematically described as follows:

$$THD = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \cdots + V_n^2}}{V_1}$$

where:

- V_1: RMS amplitude of the fundamental frequency, i.e. the first harmonic
- $V_2, V_3, V_4, \ldots$: RMS amplitudes of the harmonic frequencies

The interpretation of THD values is as follows:

- THD $< 1\%$ $\Rightarrow$ Excellent signal quality, low distortion
- THD $\approx 1\%$–5% $\Rightarrow$ Acceptable for many applications
- THD $> 10\%$ $\Rightarrow$ Significant distortion, low signal quality

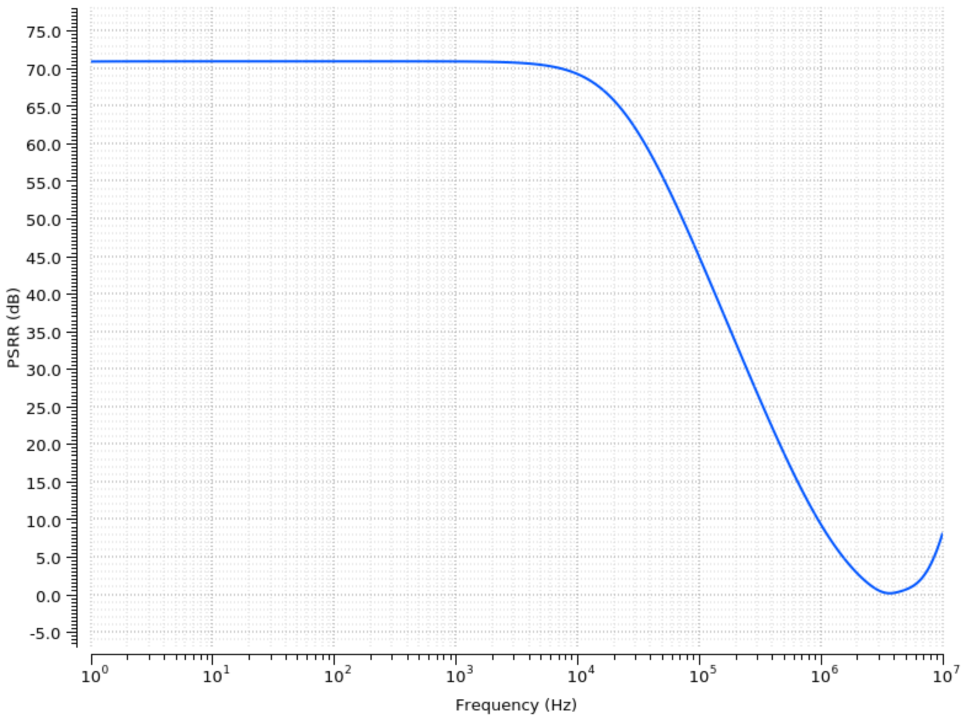


Figure 2.6: Power-supply noise rejection ratio

Although it is not easy to determine the minimum detectable amount of harmonic distortion, since various factors are involved, its minimum detectable value is selected as 1%. Therefore, the aim of the present work is to keep the harmonic distortion below 1% throughout the critical frequency range, namely $100Hz - 9kHz$.

For the THD simulations, i.e. transient analysis, different input values were tested at different frequencies. In the mid-band gain frequency range, the amplifier exhibited harmonic distortion below 1% for input voltage signals below $13.35mVpp$. Indicative THD values in the critical frequency range for different input voltages are presented in Table 2.1. The minimum input signal received by a cochlear implant amplifier from the microphone is approximately $500\mu Vpp$, while the maximum is approximately $10mVpp$; these values are amplified with distortion below 1%.

To tune the THD, the transistors operating as resistors in the feedback system of Figure 1.7 were carefully biased, as analyzed in previous Chapter. Furthermore, as the gain increases, the output has less voltage headroom, since its range is from 0 to $1.8Volt$. Therefore, reducing the gain improves the distortion. Based on theoretical calculations, the output voltage range of the amplifier before distortion is approximately from $200mV$ to $1.644V$ so that transistors Q_5, Q_6, Q_{12}, Q_{14} operate in saturation and subthreshold saturation, as mentioned in Chapter 1. This voltage range allows inputs up to $14mVpp$ before strong distortion occurs, which is consistent with the simulation results. Finally, the amplifier output offset voltage for a grounded input was calculated to be $0.1mV$, a value that does not affect the distortion. This value results from the output being

900.1mV in DC operation, with a 0.1mV deviation from the reference voltage. It should be noted that the slew-rate specification, i.e. the amplifier's ability to respond to fast signal transitions, is not relevant to designs such as the present one, because the signals do not require fast high-frequency transient behavior.

Table 2.1: Indicative THD values for different input signals

Frequency (Hz)	Input voltage (mV)	THD (%)
550	5	0.18
1200	10	0.21
1500	2	0.19
1900	0.5	0.1
2400	3.5	0.23
3300	13.35	0.94

2.2.7 Study Under Extreme Conditions

By performing the PVT analysis, the consistency of the design was evaluated against various process variations, extreme conditions, and different temperatures. MOSFETs exhibit deviations in their parameters. Many parameters of MOS transistors vary with temperature, such as the threshold voltage, the intrinsic carrier concentration of silicon, the bandgap energy, and the carrier mobility. The analysis was carried out for different temperatures and for the four extreme-condition corners, which are:

- FF (Fast-Fast) – Fast NMOS, Fast PMOS
- FS (Fast-Slow) – Fast NMOS, Slow PMOS
- SF (Slow-Fast) – Slow NMOS, Fast PMOS
- SS (Slow-Slow) – Slow NMOS, Slow PMOS

The results are summarized in Tables 2.2 and 2.3. The input-noise specification is shown in Figures 2.7 and 2.8, referring to the different corners and different temperatures, respectively. To achieve stable operation across different corners, the transistors were designed so as to reduce asymmetries between them, as analyzed in Chapter 1. It should be noted that the PSRR remains stable over the entire desired bandwidth for the various corners and temperatures.

In Table 2.2, small variations are observed in the gain and CMRR parameters. In addition, for the SS, SF, FS corners, the variations in amplifier consumption remain very close to the value under nominal conditions (1.54μW). Regarding noise in these three corners, noticeable variations are observed, but they are not unacceptable. The noise variations are mainly due to flicker noise (Figure 2.7), whose constant K varies across the different corners. The FF corner appears to have the worst performance across all specifications. In this corner, the transistor threshold voltage decreases and the carrier mobility increases. These effects increase leakage currents and transistor asymmetry, which degrade the PSRR and CMRR and increase power consumption. The main weakness in this corner appears to be noise, which increases from $196.39nV/\sqrt{Hz}$ to $404.88nV/\sqrt{Hz}$, a fairly large variation that is not ideal. The consumption of the bias circuit remains very close to that under nominal conditions ($1.44\mu W$) for all corners, with a maximum deviation of $0.05\mu W$ and minimum and maximum consumption values of $1.39\mu W$ and $1.46\mu W$, respectively.

Table 2.2: Parameter values for the different corners

Parameters	FF	SS	SF	FS
Gain (dB)	40.17	40.24	40.23	40.22
Power (Amplifier) (μW)	1.68	1.52	1.6	1.48
Noise ($nV/\sqrt{Hz}$) at $100Hz$	404.88	174.73	232.05	178.45
$PSRR(dB)$	60.6	82.06	66.6	75.69
$CMRR(dB)$ at $100Hz$	104.67	107.55	107.04	107.57

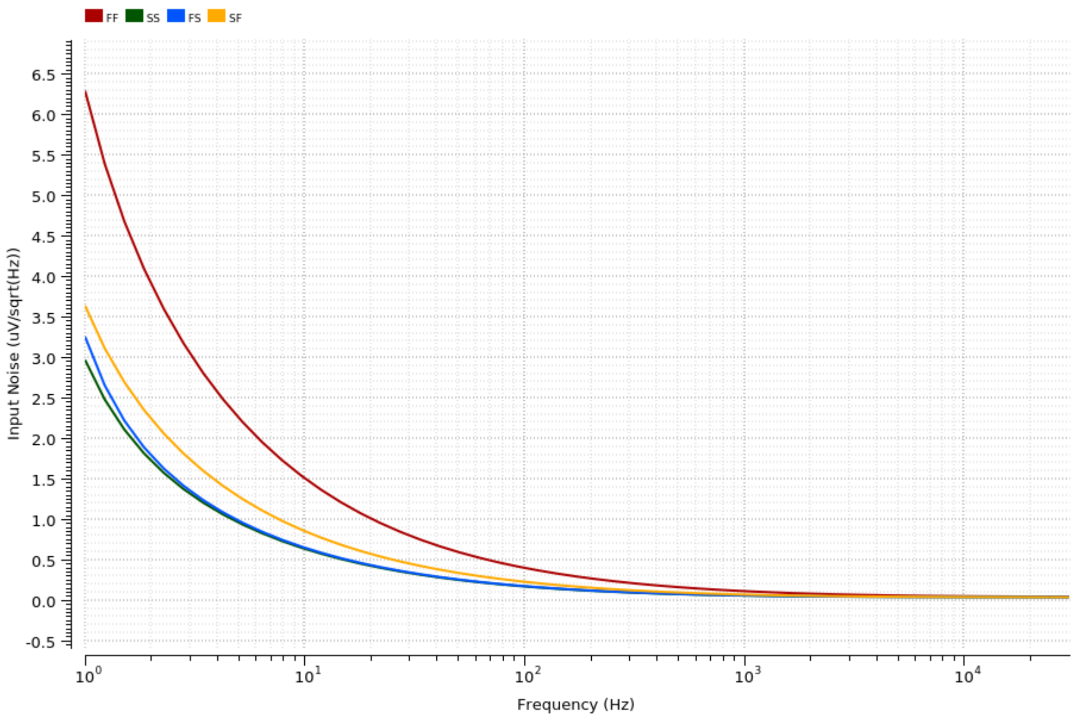


Figure 2.7: Noise for different corners

2.2.8 Study at Different Temperatures

In Table 2.3, the amplifier shows minimal variation in the gain and CMRR parameters. The PSRR decreases by approximately $5dB$ from its nominal value at $45°C$. This occurs because, as temperature increases, carrier mobility decreases, reducing the transconductance and therefore the PSRR. The PSRR remains slightly above $65dB$ and is therefore at a satisfactory level. As expected, power decreases as temperature decreases and increases as temperature increases, mainly due to the corresponding decrease or increase in leakage currents. At $45°C$, power increases by approximately $0.07\mu W$ and remains at satisfactory levels. In addition, at low temperatures thermal noise decreases and thus the total noise decreases as well, while the opposite occurs at high temperatures. Specifically, the noise at $45°C$ shows a noticeable increase from $196.39nV/\sqrt{Hz}$, although it is not unacceptable. It should be noted that, as shown in Figure 2.8, the noise increase is not based so much on thermal noise but on flicker noise, which is positive because the total noise density over the $20 - 20kHz$ spectrum is degraded only at the lower frequencies. Finally, harmonic distortion remains below 1% over the desired input-signal range ($500\mu V - 10mV$) for all corners and temperatures.

Table 2.3: Parameter values for temperature variations

Parameters	-30°C	0°C	45°C
Gain (dB)	40.26	40.25	40.19
Power (Amplifier) (μW)	1.46	1.49	1.61
Noise ($nV/\sqrt{Hz}$) at $100Hz$	150	162.44	269.43
$PSRR(dB)$	95.42	80.95	65.79
$CMRR(dB)$ at $100Hz$	108.7	108.17	106.19

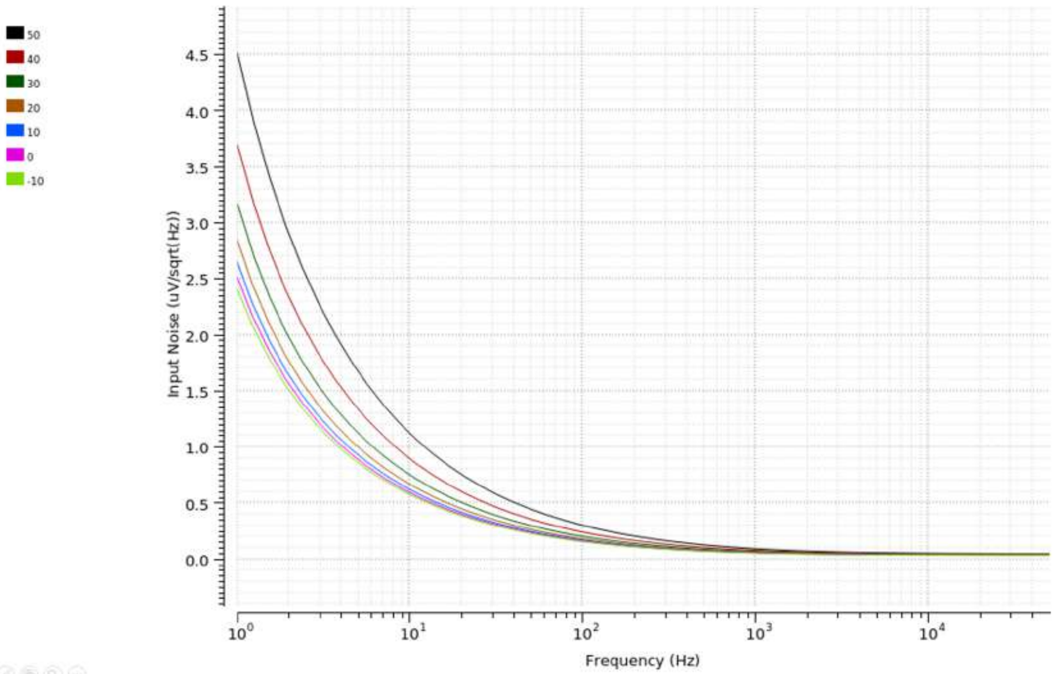


Figure 2.8: Noise for different temperatures

2.2.9 Layout Results

Figure 2.9 shows the final physical design in analog extraction form of the amplifier, the feedback system together with the output high-pass filter, and the bias circuit. The total occupied area is $294.475\mu m \times 135.56\mu m \approx 0.04mm^2$. More specifically, the dimensions of the amplifier without the feedback system were measured to be approximately $129\mu m \times 74\mu m$, while those of the bias circuit without the resistor were approximately $80\mu m \times 52\mu m$. The resistor of the bias circuit was placed at the bottom of the design and connected to the bias circuit using metal 4. This is one of the few cases where metal 4 had to be used. The area occupied by the resistor is approximately $115\mu m \times 2\mu m$. As can also be seen from Figure 2.9, most of the area is occupied by the input capacitors, located on the left side ($131\mu m \times 131\mu m$). All capacitors used are METAL-INSULATOR-METAL (MIM) capacitors. The bias circuit is shown in the upper-right part, while the capacitor of the high-pass filter with a multiplier equal to 4 is shown in the lower-right part. To its left is the capacitor at the amplifier output.

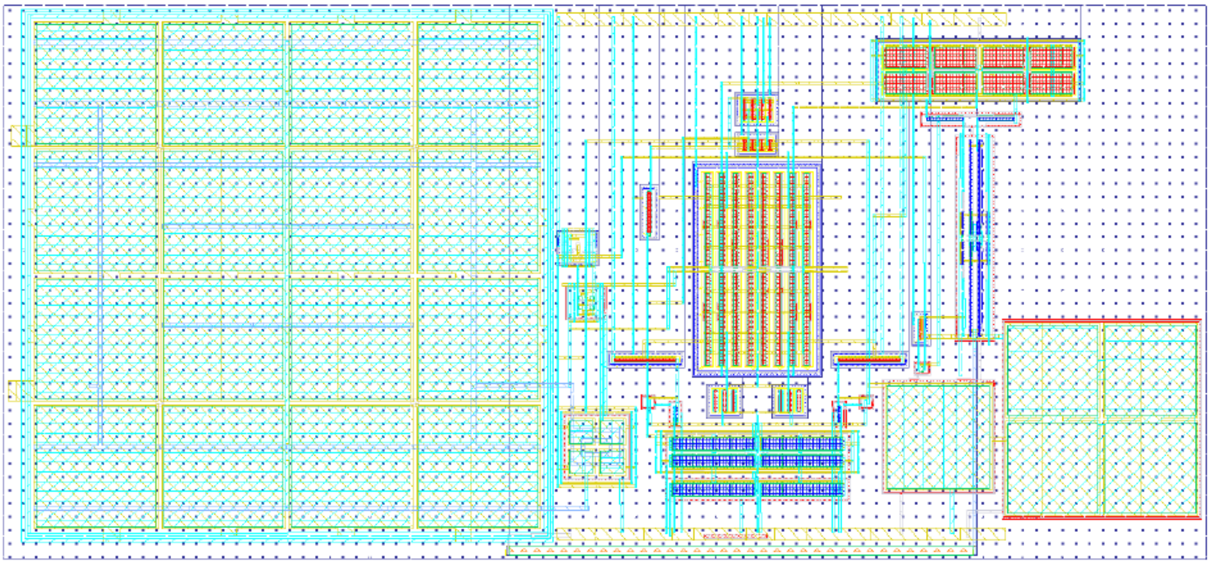


Figure 2.9: Physical design: Amplifier–feedback system–output filter–bias circuit

The design methodologies and techniques are analyzed in Chapter 1 and were applied to all transistors and capacitors. More specifically, one example is the capacitors C_1 and C_2, each of which uses a multiplier equal to 8 and was placed using a common centroid technique, with capacitor C_1 arranged in a crossed configuration, four elements to the left and four to the right. The remaining capacitors form C_2. In this way, their variations cancel each other out.

The results of the physical-design analysis are shown in Figures 2.10 and 2.11 for the noise and CMRR specifications, and in Table 2.4 for all specifications together. Harmonic distortion remains below 1% over the entire desired input-signal range ($500\mu V - 10mV$). As shown in Figure 2.10, although the flicker noise is increased, within the desired frequency range ($20Hz - 20kHz$) it remains very close to the schematic-level simulations. The power, gain, and PSRR specifications are also at similar levels. The only specification that shows a significant deviation is the CMRR. As shown in Figure 2.11, the CMRR is slightly above $95dB$ at low frequencies and begins to coincide with the schematic result at approximately $1kHz$. The CMRR is mainly due to the matching of the input transistors and capacitors $C_1 - C_2, C_3 - C_4$. Different common centroid arrangements were used, and the one with results closest to the schematic was selected.

Table 2.4: Specification values for physical-design analysis

Parameters	Schematic	Physical design
Gain (dB)	40.26	40.22
Power (Amplifier+Bias circuit) (μW)	2.98	2.99
Noise ($nV/\sqrt{Hz}$) at $100Hz$	196.39	199.98
RMS noise (μV) ($20Hz - 20kHz$)	7.41	7.58
$PSRR(dB)$	71	70.22
$CMRR(dB)$ at $100Hz$	107.36	95.26

Table 2.5 shows the physical-design results for the different corners. The resulting values are interpreted in a similar way to the schematic-level results described in the previous section.

Table 2.5: Physical-design specification values for the different corners

Parameters	FF	SS	SF	FS
Power (Amplifier+Bias circuit) (μW)	3.05	3.02	3.04	2.92
Noise ($nV/\sqrt{Hz}$) at $100Hz$	433.9	177.7	235.68	182.38
$PSRR(dB)$	59.87	76.18	64.96	71.1
$CMRR(dB)$ at $100Hz$	93.89	95.63	94.94	95.48

2.3 Application of the Reference Voltage Circuit

Until now, a DC voltage source of $900mV$ has been used to bias the amplifier input. In practice, this voltage must be generated by a reference voltage circuit. Therefore, the purpose of the reference voltage circuit is to provide a stable input bias voltage for the amplifier that is independent of temperature and different corners; the latter has not been achieved.

As explained in Chapter 1, the bias-current value is given by $I_{sub} = \frac{V_{GS31} - V_{GS30}}{R_1}$. Taking the current equations of transistors Q_{30} and $Q31$ and dividing them side by side, since the transistors operate in the subthreshold region, we obtain $V_{GS31} - V_{GS30} = nV\tau ln(\frac{(W/L)_{30}}{(W/L)_{31}})$. Through simulations, this value was adjusted to be approximately $49mV$, and therefore the current is equal to $520nA$.

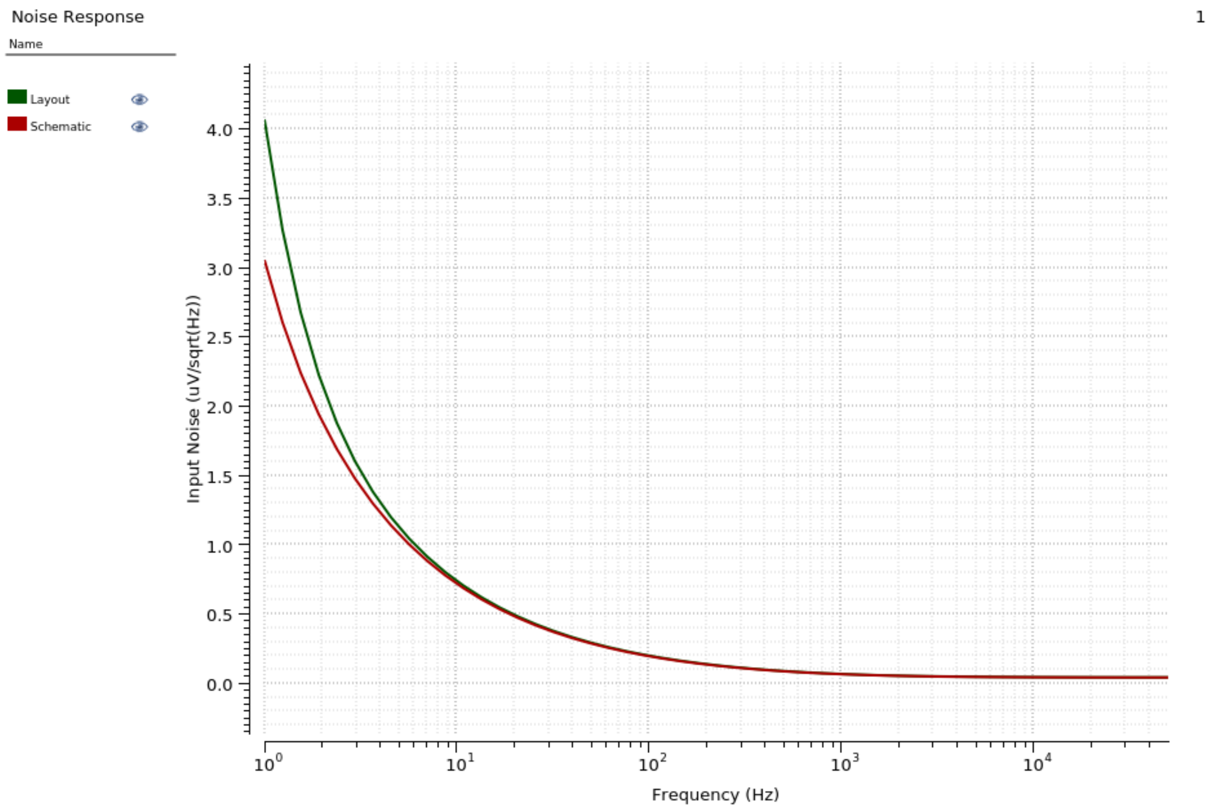


Figure 2.10: Input-referred noise for the physical design (green)

With the introduction of the reference voltage circuit, the noise and power-supply rejection specifications are affected, as analyzed in previous Chapter. The remaining specifications stay the same. The low-pass filter at the output, formed by the output resistance of the reference voltage circuit and capacitor C_7, helps reject the flicker noise added by the reference voltage circuit at the amplifier input in the desired operating range of $20 - 20kHz$. Correspondingly, the PSRR in the same operating range achieves the desired rejection of $71dB$. These results are shown in Figure 2.12. Nevertheless, although operation in the desired range is maintained, the use of the amplifier in similar applications with lower-frequency operating ranges is not possible with this reference voltage circuit design. Furthermore, as shown in Figure 2.13, over a temperature range from $-30°C$ to $50°C$, the circuit has a maximum reference-voltage deviation at a satisfactory level, approximately $2.8mV$ from $900mV$. As already mentioned, simulations of the reference voltage circuit for different corners showed significant variations in the output voltage, on the order of $25mV$ to $60mV$ from the nominal value of $900mV$. This deviation makes the circuit unsuitable for use in its current form, requiring further design and optimization to improve the stability and accuracy of the reference voltage.

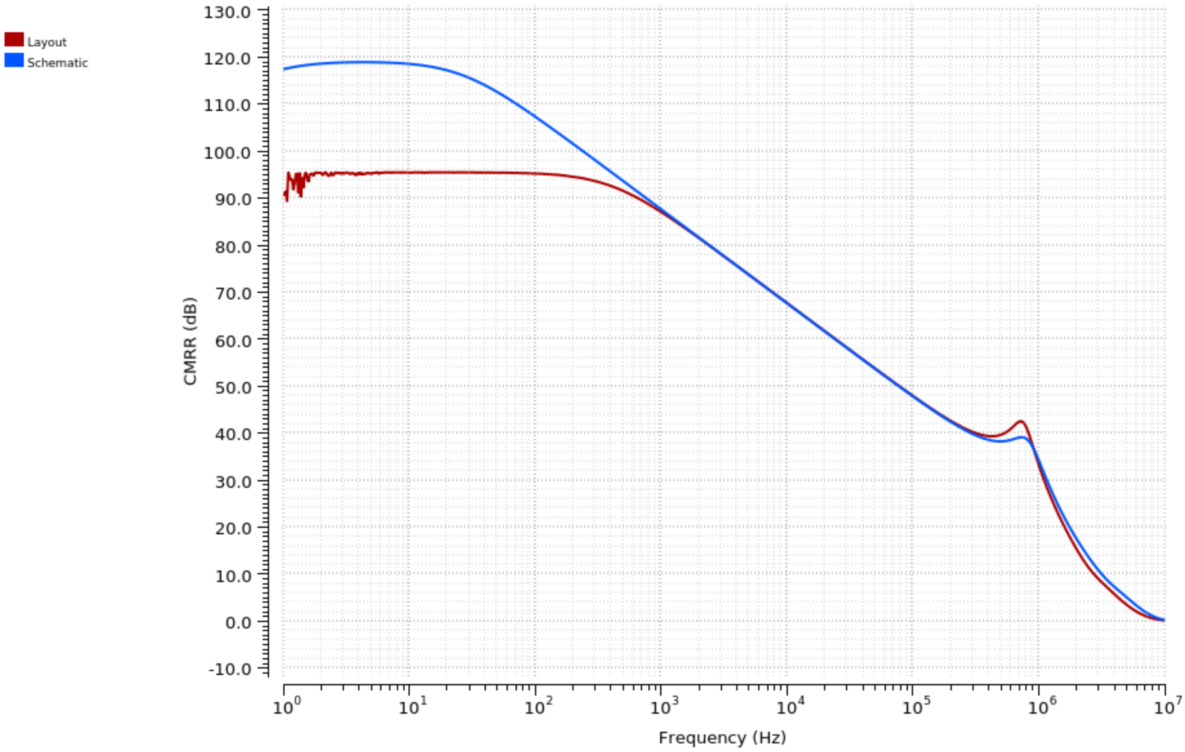


Figure 2.11: CMRR of the physical design (blue)

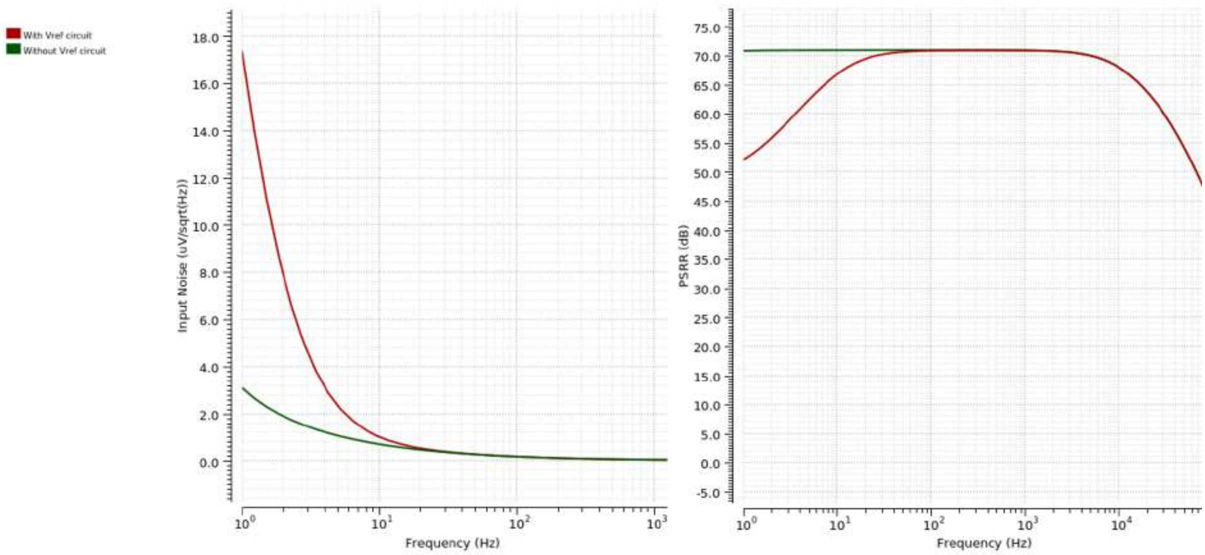


Figure 2.12: Variation of PSRR and noise when using the reference voltage circuit

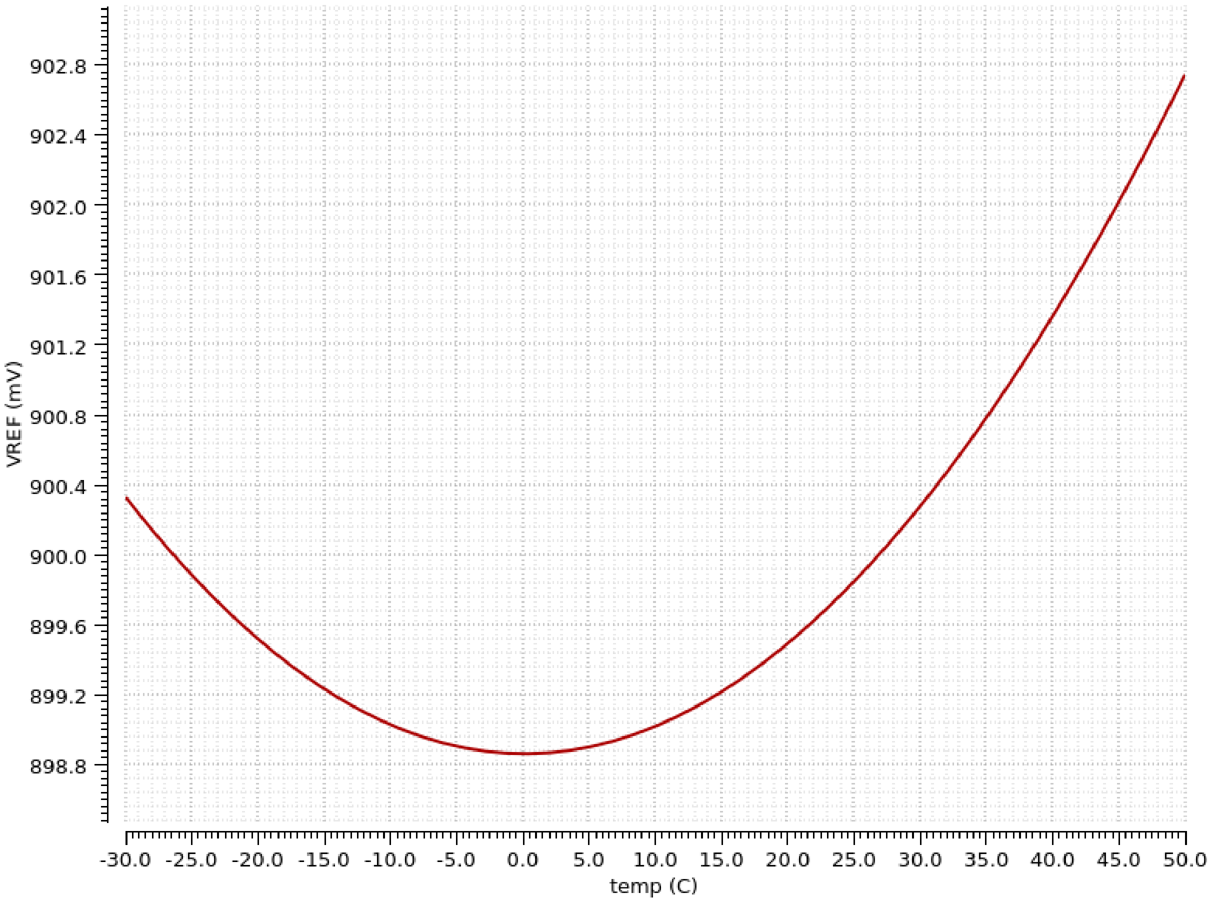


Figure 2.13: Dependence of the generated reference voltage on temperature

Chapter 3

Conclusions

In this chapter, the amplifier is compared with similar amplifier designs in terms of the overall specification values. First, however, the behavior of the amplifier for different input bias currents is compared in order to better explain the trade-off between noise and power consumption.

3.1 Trade-Off Between Noise and Consumption

Table 3.1 shows the schematic-level behavior of the amplifier for different input bias-current values. The table shows that there is a trade-off between noise and power, since when the bias current increases, power consumption becomes higher, while the noise decreases at the same time. This occurs because the transconductance of the input transistors, and consequently of the amplifier, increases, which reduces the total noise contribution of the transistors, as described in Chapter 1. The CMRR and PSRR specifications improve with the increase in transconductance and, consequently, in input bias current. This happens because although the amplifier differential gain remains constant due to closed-loop operation, the gains from the supply and from the common-mode signal decrease as transconductance increases.

Table 3.1: Parameter values for different bias-current values

Parameters	$300nA$	$400nA$	$500nA$
Power (μW)	1.02	1.54	2.07
Input noise ($nV/\sqrt{Hz}$) at $100Hz$	242.23	196.39	178.98
Input noise (μV) ($1-10kHz$)	9	7.44	6.75
$PSRR(dB)$	66.55	71	74.74
$CMRR(dB)$ at $100Hz$	104.8	107.36	109.29

3.2 Comparison with Similar Amplifiers

Table 3.2 summarizes results from amplifiers for biomedical applications. The last column corresponds to the amplifier of the present work, using the post-layout analysis results.

Power and area consumption refer only to the amplifier, for a clearer comparison with the rest. Most amplifiers, and specifically those used for cochlear implants [15] [7], have a gain of around $40dB$. The frequency bandwidths differ depending on the application. For cochlear implants, the bandwidth is approximately $20 - 20kHz$.

As shown, the power-supply rejection and common-mode rejection specifications range from $60 - 84dB$ and $62.9 - 104dB$, respectively. Therefore, the amplifier of this design is within both specification ranges. An improvement could be made so that the CMRR exhibits less variation over the amplifier operating range.

Regarding area consumption, the amplifier is within the desired limits. The reduced area consumption is related to the use of low input-capacitance values equal to $7pF$ and input differential-pair transistors with a smaller aspect ratio compared with other designs, at the cost of increased noise. The somewhat increased area consumption in [10] and [19] is due to the use of six source degeneration resistors of $186k\Omega$ and input capacitors on the order of $20pF$, while the considerably reduced area in [20] is due to the use of off-chip input capacitors. The amplifier of this design achieves fairly low consumption compared with others, but this is done at the expense of the noise specification, which increases to some extent.

The reason the compromise between lower consumption and increased noise was selected is related to the fact that, in cochlear implants, a large part of the flicker noise is removed because of the operating bandwidth. In particular, in the critical operating range, approximately $500Hz - 9kHz$, thermal noise dominates and remains at low levels. Thus, sacrificing a small portion of the noise specification was considered acceptable in order to achieve lower consumption. This is also reflected in the cochlear-implant amplifiers in the related table [15] [7], where the noise specifications are more relaxed. In contrast, in amplifiers with lower-frequency operating ranges, stricter noise specifications are required so that signals are not significantly degraded by increased flicker noise.

In conclusion, as explained in Table 3.1, there is a trade-off between power consumption and noise. Examples of higher consumption for lower noise are found in [10] [21] [3], while the opposite is observed in [15] [22]. The present work belongs to the second category. The best balance between noise and power consumption is achieved by [20] [19]. To adapt the amplifier of the present work to other applications such as brain-computer interfaces, neural-activity monitoring, electrocardiography, and biosignal amplifiers in general, the frequency range must be reduced and the power consumption must be increased in order to reduce noise. At the same time, additional tuning must be performed so that the remaining specifications are maintained without negative effects.

Table 3.2: Comparison of different amplifiers

Specifications	[20]	[15]	[10]	[21]	[23]	[24]	[22]	[3]	[6]	[7]	[19]	Present Work
Technology (nm)	180	180	180	180	180	180	180	180	180	180	180	180
Supply (V)	1	1	1.8	1.8	1.8	1.4	1	1.8	1.8	1.8	1	1.8
Gain (dB)	51.06	43.75	46/40	20	39.7	39.22	38.06	39.75	61.52	43.7	60/54	40.22
Bandwidth (Hz)	$378m$ - 619	22.55–$18.85k$	0.54–6.1k	100- $10k$	0.36–$11.8k$	$3 - 5k$	1 - 137	0.3– $4.4k$	$13.4k$	18.7–$20.6k$	$1 - 10k$	27.12 - $19.8k$
Power (μW)	1.62	1.69	6.84	5.76	2.25	2.98	0.0502	4.07	3.345	4.6	1	1.55
Input noise ($V/\sqrt{Hz}$)	$918n$ @ $1Hz$	2.71μ @ $1Hz$, $657n$ @ $20Hz$, $90.1n$ @ $20kHz$	$30.1n$ @ $1kHz$	NR	NR	NR	9μ @ $1Hz$, 2μ @ $20Hz$, 1μ @ $100Hz$	NR	NR	$473.47n$ @ $4kHz$	$20n$ @ $10kHz$	4.06μ @ $1Hz$, $491.49n$ @ $20Hz$, $48.69n$ @ $4kHz$, $44.82n$ @ $10kHz$
Input noise ($\mu Vrms$)	2.4	NR	3.1 ($1Hz$–$6.1kHz$)	1	4.04	3.03 ($1Hz$–$10kHz$)	NR	3.19 ($1Hz$–$10kHz$)	3.3	NR	4.2 ($1Hz$–$10kHz$)	7.84 ($1Hz$–$10kHz$)
PSRR (dB)	NR	NR	84	60	64.6	77	NR	77.6	84	NR	NR	70.22
CMRR (dB)	97.4	86.11	66	NR	62.9	70	96.17	76	104.52	NR	110	62-95
Area (mm^2)	0.0032	NR	0.082	NR	NR	0.046	NR	0.058	NR	NR	0.09	0.036

3.3 Suggestions for Future Research–Improvements

In conclusion, the conventional folded cascode operational amplifier alone would not be able to meet the requirements of the application. To satisfy the required specifications, additional techniques must be applied to the conventional folded cascode operational amplifier, such as composite cascode, conventional cascode, flipped current follower, active cascode, and self-biasing of the output cascode current mirror.

Under nominal conditions, the amplifier operates within the desired limits, satisfying the specifications for noise, power consumption, phase margin, gain, bandwidth, harmonic distortion, area, power-supply rejection, and common-mode rejection. These characteristics make it capable of amplifying the weak signals received from the microphone under typical operating conditions.

The amplifier has room for improvement regarding its operation over a wide temperature range and under extreme conditions. In these cases, variations are observed in the noise, power-consumption, and power-supply-rejection parameters. The lowest amplifier performance is observed in the FF (Fast NMOS Fast PMOS) corner, where the power-supply rejection ratio decreases by approximately $10dB$, while the power consumption increases by $0.14\mu W$ relative to the nominal-condition values. The most negative effect in this corner concerns noise, which becomes slightly more than twice as large over a large part of the frequency spectrum within the operating range.

In addition, the amplifier has room for improvement in its physical design. Specifically, the flicker noise is slightly increased, although this is not a significant issue for the present application, since most signals at these frequencies do not lie within the operating spectrum. The main weakness of the physical design is that the common-mode rejection ratio (CMRR) is degraded up to frequencies of $1kHz$, to approximately $95dB$, and does not achieve the value obtained at schematic level.

Finally, there is always room for future research into improving the balance between noise and power consumption, with the aim of designing an even more efficient amplifier.

Appendix A

The results presented in this work were produced using the Computing Cluster and the support services provided by the Information Technology Center of Aristotle University of Thessaloniki. The authors would like to thank the Information Technology Center of Aristotle University of Thessaloniki for its support during this research work.

The design and simulations were carried out using the X-FAB 180 nm CMOS technology within the Cadence Virtuoso design environment. At the schematic level, the required MOS devices and passive components were instantiated from the technology-provided device libraries. The selected capacitors and resistors correspond to MIM-capacitor and diffusion-resistor available in the process.

After the schematic was designed and the pins and the amplifier symbol were created through `Create->Cellview->From Cellview`, a test bench was created in order to perform the simulations. Several different schematics were created so that all the different designed circuits could be tested both separately and together. For the schematic-level simulations, two tools were used: ADE L and ADE Explorer. In ADE L, the simulation is selected through the `Analyses->Choose...` window. The simulation temperature is selected from the ADE L menu in the upper left corner. In addition, before running the simulations, the files required for X-FAB are installed through `Setup->Model Libraries`. The `ac` source `vsin` from `analogLib` was placed at the input of the feedback system, together with the corresponding `dc` source for the supply. All the pins of the amplifier symbol were connected, and the simulations for the specifications were then performed.

For the gain-related specifications, namely gain, CMRR and PSRR, as well as for the phase margin and frequency bandwidth, an `ac` analysis was performed, as shown in Figure 1. The frequency `Sweep Range` was defined and the `Sweep Type` was set to logarithmic, with 100 points per decade. The results are displayed through `Results->Direct Plot->AC Gain & Phase`, by first selecting the reference output and then the input.

For the noise specification, a `noise` analysis was performed, as shown in Figure 2. The `Sweep Range` was defined, and in the `Output Noise` and `Input Noise` fields, the corresponding node and input voltage source were selected. The results are displayed through `Results->Direct Plot->Equivalent Input Noise`.

For all DC operating points, a `dc` analysis was performed and the `Save DC Operating Point` option was enabled, as shown in Figure 3. To display the operating points on the

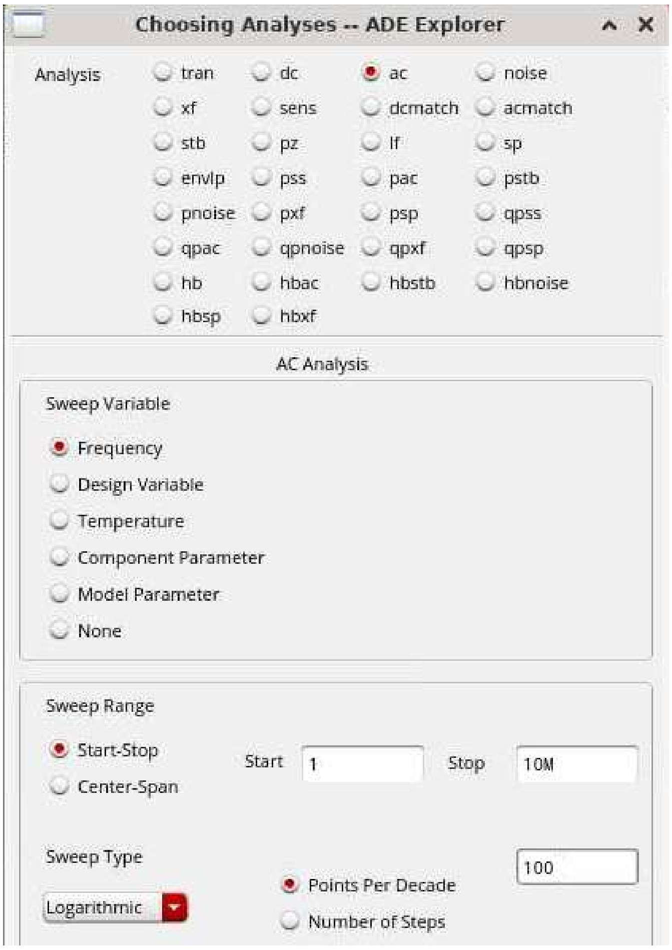


Figure 1: `ac` analysis

schematic, the option `right click->Annotations->DC Operating Points` was selected. Some operating points then appear on the schematic. The selection of which operating points are displayed is made through `right click->Annotations->Setup...`.

For the harmonic distortion specification, different frequency values, for example 1 kHz, and different signal amplitudes, for example $10\,\mathrm{mV_{pp}}$, were selected for the input `ac` source, using all possible combinations. A `tran` analysis was then performed, as shown in Figure 4, with the `Stop Time` selected according to the input frequency, so that a sufficiently large sample could be obtained. The `conservative` option was also selected for higher simulation accuracy. Then, through `Outputs->To Be Plotted->Select On Design`, the output to be simulated was selected, namely the output of the amplifier and the filter. After running the simulation, the `Measurements->Spectrum` option was selected in the window that opened, and 10 harmonics were chosen. By selecting `Plot`, the total harmonic distortion (THD) is displayed.

Figure 5 shows that the series connection of diode-connected NMOS and PMOS transistors in the feedback system improves the harmonic distortion compared with the use of only NMOS or only PMOS transistors. More specifically, it illustrates how the distortion produced by one device cancels that produced by the other. For this simulation, an arbi-

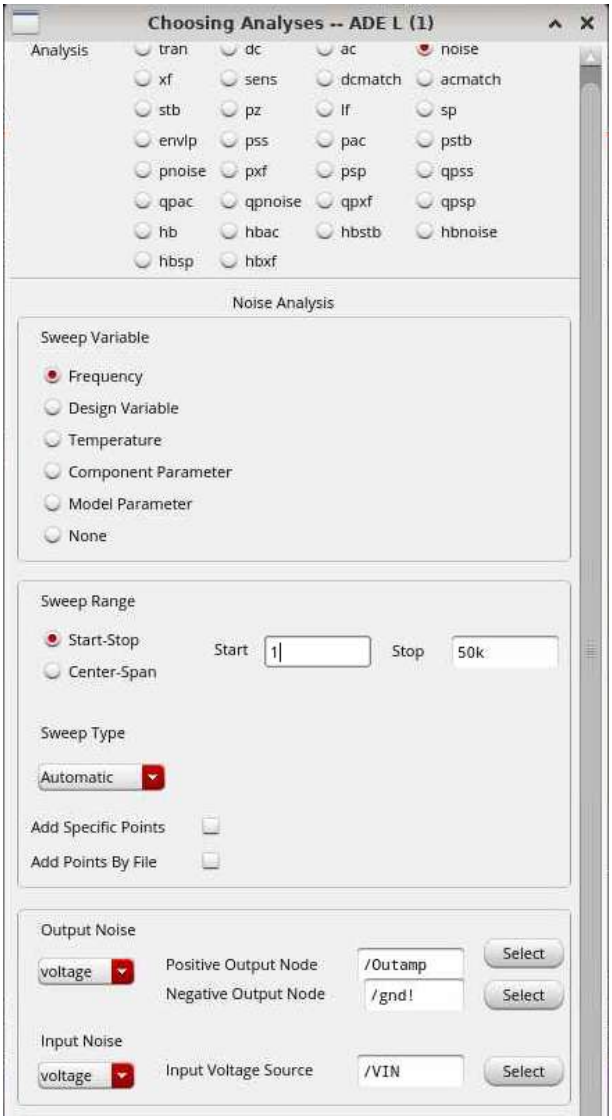


Figure 2: `noise` analysis

trary input signal of 10.4 mV at a frequency of 1 kHz was used. The `outamp` signal refers to the signal at the amplifier output for the case where NMOS and PMOS transistors are connected in series. The measured THD is approximately equal to 0.19%.

ADE Explorer and the Maestro tool were used for corner analysis, as shown in Figure ??, since they are able to run multiple simulations in parallel. By selecting `Corners`, a new window opens, where the `Add New Corner` icon at the top is selected in order to add a new corner. On the left side, under `Model Files`, the X-FAB files are installed in the same way as in ADE L. The corners are the following:

- `WP = Worst case power = Fast NMOS & Fast PMOS`
- `WS = Worst case speed = Slow NMOS & Slow PMOS`
- `WO = Worst case one = Fast NMOS & Slow PMOS`
- `WZ = Worst case zero = Slow NMOS & Fast PMOS`

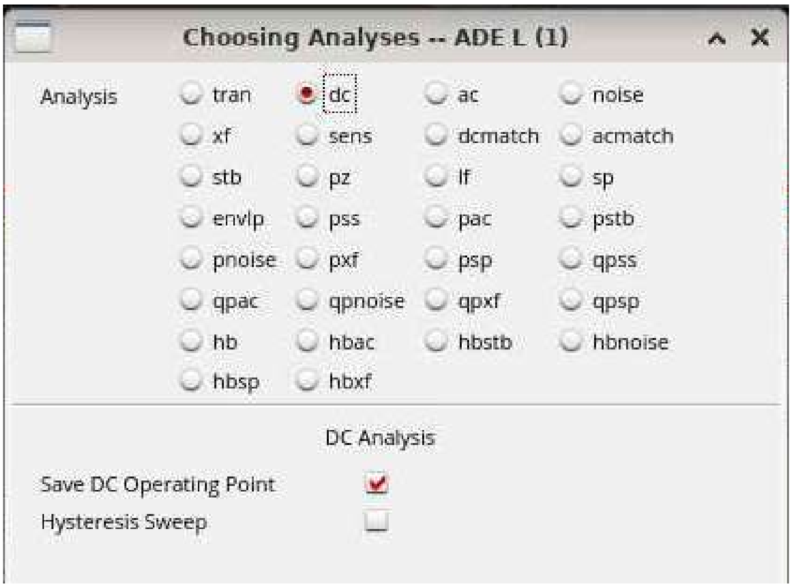


Figure 3: `dc` analysis

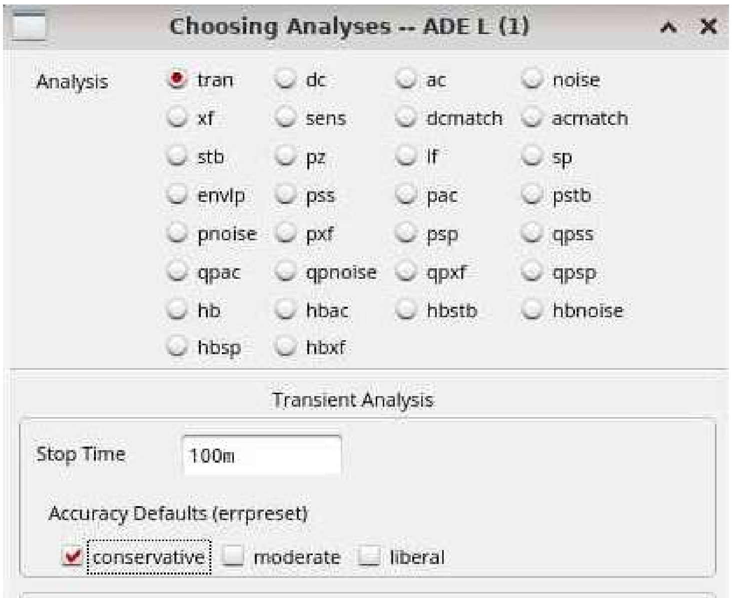


Figure 4: `tran` analysis

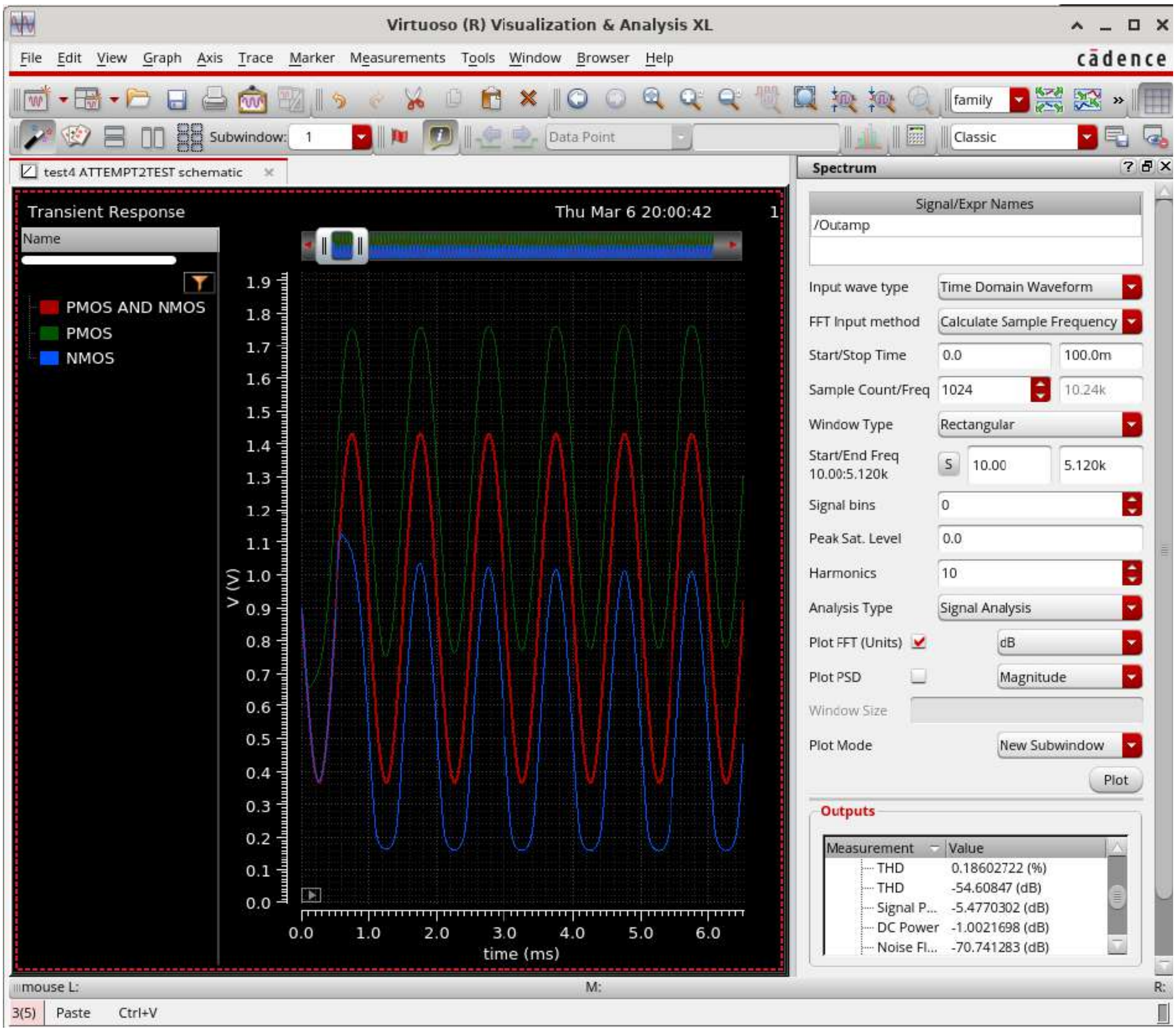


Figure 5: THD improvement using NMOS and PMOS transistors in series in the feedback system, shown in red

Appendix B

In order to create the layout, all elements to be designed must be included in the same project. Initially, the amplifier layout was designed, followed by the feedback circuit, the output filter and the biasing circuit. In this way, by performing post-layout analysis for each part separately, possible problems could be identified more easily. The individual parts were then combined. Several layout versions with minor variations were created, mainly concerning transistor and capacitor matching. The optimal topology was then selected.

To move to the Layout XL environment, where the physical design was carried out, the option `Launch->Layout XL` is selected, opening a new window for the layout design. In the new window, by selecting `Generate All From Source` at the lower right, all transistors, capacitors, resistors and pins of the schematic appear in Layout XL. The left-hand window shows all available layers. The technology provides metal layers from Metal 1 up to Metal 4. The design is performed through the `Create` window, where the desired element is selected, such as a wire through `Wiring->Wire`, a via, or an MPP guard ring.

During physical design, checks are performed to verify whether the layout satisfies the design rules of the technology. Due to a technical issue in the X-FAB installation, the DRC (Design Rule Check) is performed through `Assura->Run DRC...`, since the `PVS->Run DRC...` tool is not fully reliable. Figure 6 shows the message displayed when the design rules are satisfied. When all connections have been completed, the layout is checked against the schematic through `Assura->Run LVS...`, and the corresponding message is displayed, as shown in Figure 7. If some connections do not match, they can be located through `Window->Assistants->Annotation Browser`.

Figure 6: Message displayed when the design rules are satisfied

After the physical design has been completed, no DRC errors are present, and the schematic and layout match, analog extraction is performed for post-layout analysis. For

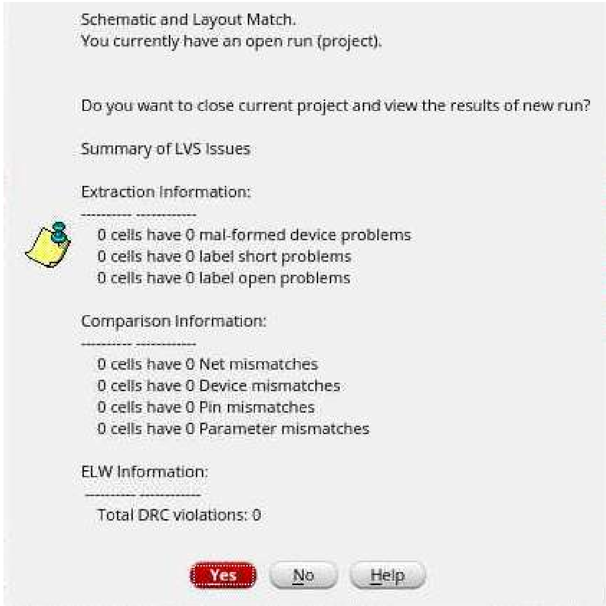


Figure 7: Message displayed when the layout matches the schematic

this purpose, `Assura->Run Quantus` is selected and configured as shown in Figure 8. Then, a `config`-type test bench is created, which is used for post-layout analysis. Through the Hierarchy Editor window, shown in Figure 9, which opens together with the test bench, the desired view of the cell is selected, either `analog_extracted` or `schematic`, by right-clicking on the cell and selecting `Set Cell View`. The simulations are then run in the same way as for the schematic. By selecting `Append` in the ADE L environment, it is possible to switch between `analog_extracted` and `schematic` while preserving the simulation results, allowing for a more direct comparison.

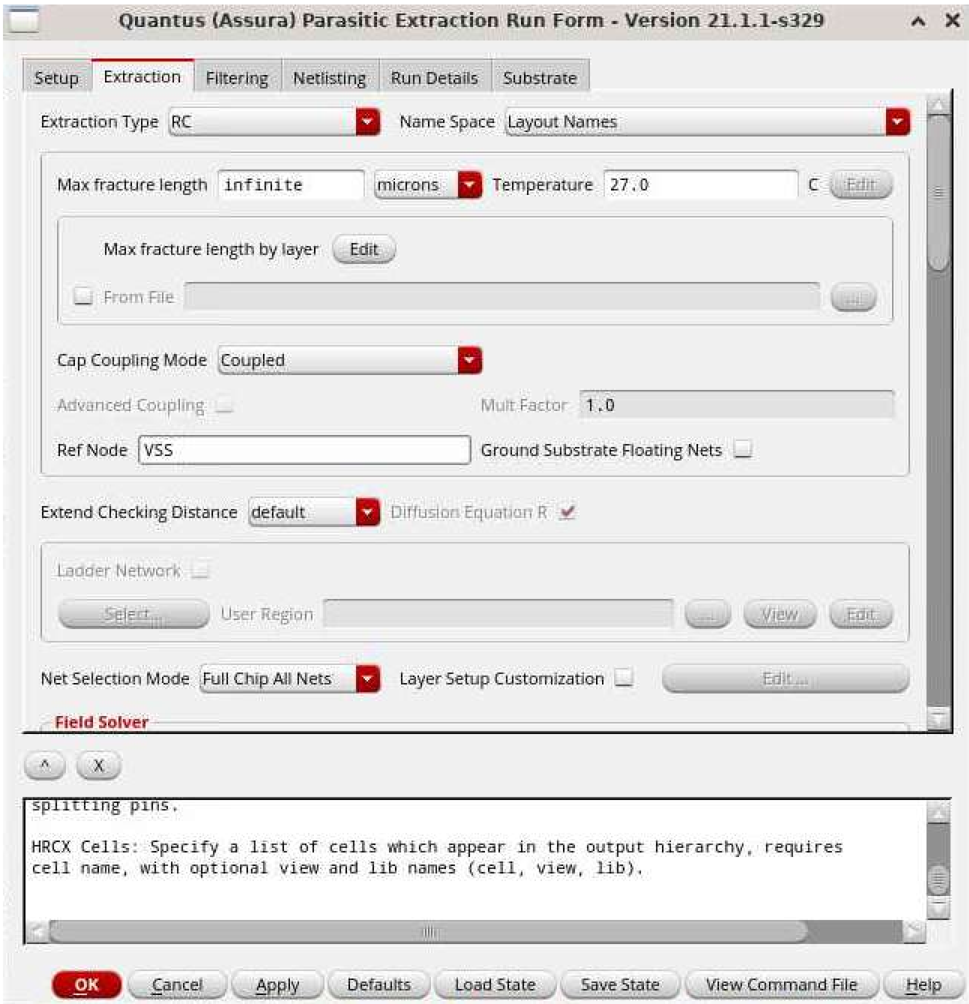
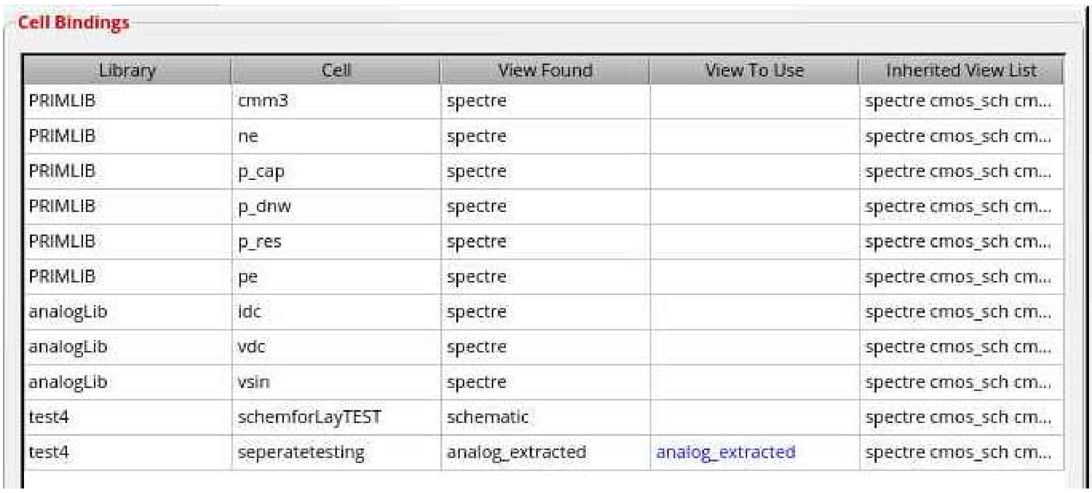


Figure 8: Quantus window



Cell Bindings

Library	Cell	View Found	View To Use	Inherited View List
PRIMLIB	cmm3	spectre		spectre cmos_sch cm...
PRIMLIB	ne	spectre		spectre cmos_sch cm...
PRIMLIB	p_cap	spectre		spectre cmos_sch cm...
PRIMLIB	p_dnw	spectre		spectre cmos_sch cm...
PRIMLIB	p_res	spectre		spectre cmos_sch cm...
PRIMLIB	pe	spectre		spectre cmos_sch cm...
analogLib	idc	spectre		spectre cmos_sch cm...
analogLib	vdc	spectre		spectre cmos_sch cm...
analogLib	vsin	spectre		spectre cmos_sch cm...
test4	schemforLayTEST	schematic		spectre cmos_sch cm...
test4	seperatetesting	analog_extracted	analog_extracted	spectre cmos_sch cm...

Figure 9: Hierarchy Editor

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